

# Models, calibration and inference

The construction of a volatility surface from option quotes, and its evolution between observations.

### Market inputs

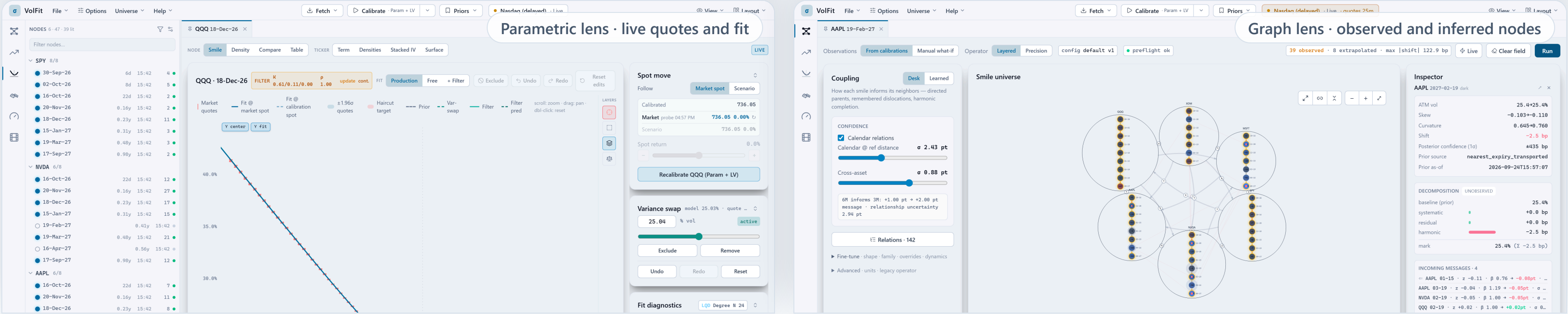
Forwards, dividends, exercise style, quote timestamps and bid–ask information.

### Calibration

Smile models, local volatility, the objective, tail assumptions and consistency across maturities.

### Subsequent updates

Event time, spot transport, prior persistence, temporal filtering and graph inference.



# Objects and calculation order

A node is one underlying and one expiry. A surface couples the nodes belonging to an underlying.

| STAGE     | INPUT                                 | OUTPUT                                      |
|-----------|---------------------------------------|---------------------------------------------|
| Prepare   | Quotes, exercise style, spot, carry   | European prices, F, D and total variance    |
| Fit       | Prepared quotes + objective + model   | A smile, or a local-variance surface        |
| Constrain | Adjacent expiries and tail policy     | Calendar diagnostics and a repaired surface |
| Update    | Spot move and saved state             | Transported fit, prior or filtered estimate |
| Infer     | Observed handle changes and relations | Dark-node handles, smile and uncertainty    |

## NOTATION AND UNITS

$x = K/F$ ;  $k = \log(K/F)$ ;  $c = C/(DF)$ ;  $w = \sigma^2\tau$ . Calendar time  $t$  drives carry and exercise. The variance clock  $\tau$  defines the volatility reading. One vol bp is 0.0001 of absolute volatility.

IN THE APP

Open one node on the Parametric lens and read the Fit diagnostics card: the three handles, the RMS on the smile and on the surface.

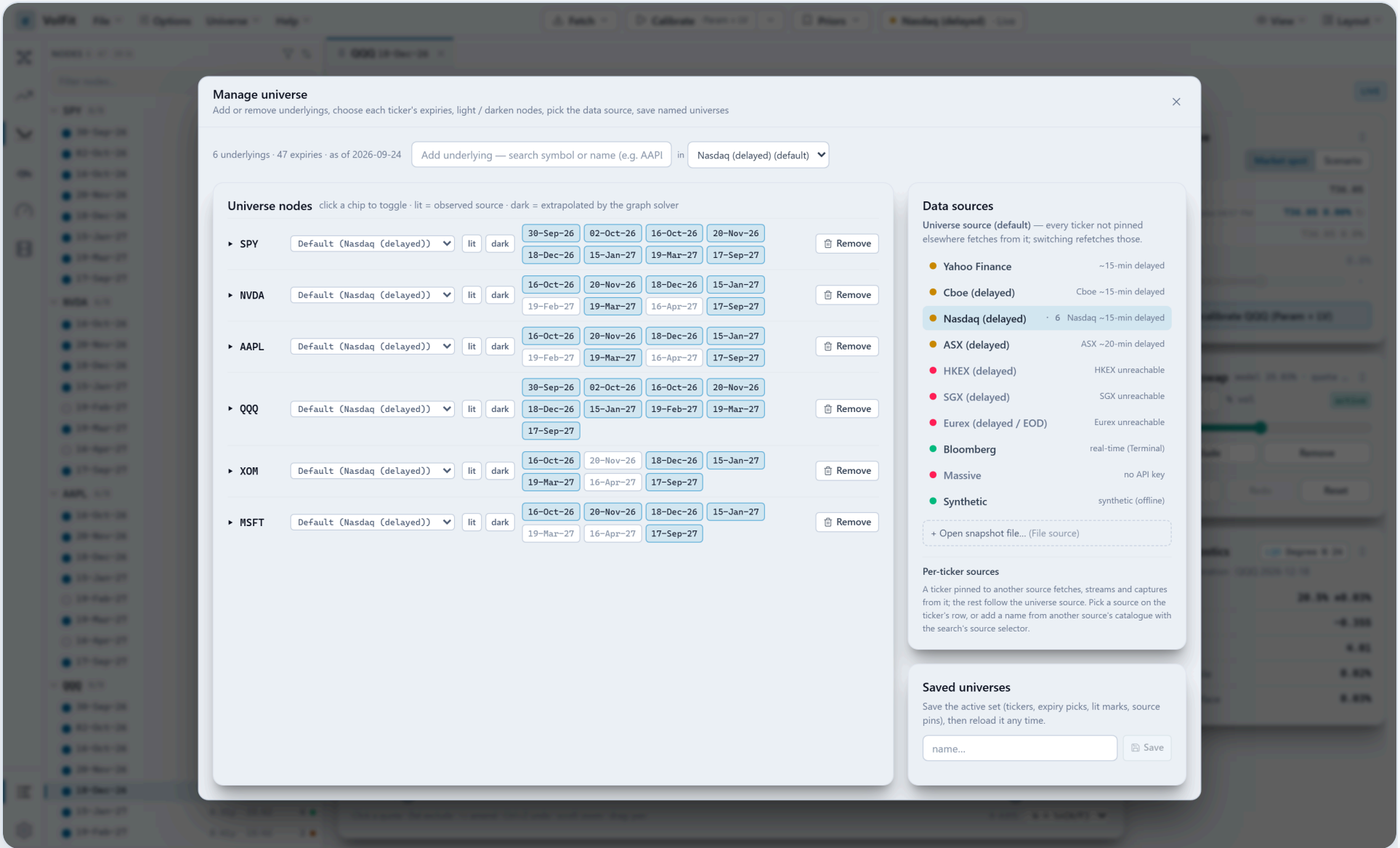
# Universe, observations and fit state

The universe selection determines which smiles are observed and which are inferred.

- **Lit nodes:** Their current quotes enter calibration. Dark nodes receive an inferred result from the graph.
- **A calibration snapshot:** The fit retains the quotes, spot, forwards and settings used to produce it. Changed inputs can make it stale.
- **Separate operations:** Fetch refreshes observations. Calibrate re-estimates parameters. Spot updates transport the stored fit.

IN THE APP

Open Manage universe, identify one lit and one dark expiry, then show the Calibrate scope menu and the stale indicator.



24 September capture: six underlyings, 47 selected expiries. Filled and outlined expiry controls identify lit and dark nodes.

A displayed chain can combine continuously updated contracts with older observations.

- **Snapshot:** A request returns the selected chain and spot. Stored historical frames are checked before requesting the same instant again.
- **Streaming:** The provider updates a local quote book. Subscription allocation favours the visible node, then distributes capacity across names.
- **Outside the live allocation:** Massive uses a periodic REST layer. Bloomberg can rotate overflow contracts through temporary subscription buckets.

## Data sources

Universe source (default) — every ticker not pinned elsewhere fetches from it; switching refetches those.

- **Yahoo Finance** ~15-min delayed
- **Cboe (delayed)** · 1 Cboe ~15-min delayed
- **Nasdaq (delayed)** Nasdaq ~15-min delayed
- **ASX (delayed)** ASX ~20-min delayed
- **HKEX (delayed)** HKEX unreachable
- **SGX (delayed)** SGX unreachable
- **Eurex (delayed / EOD)** Eurex unreachable
- **Bloomberg** real-time (Terminal)
- **Massive** · 1 delayed feed · streaming 420 · ...  
420 acked · 98 msg/s · last tick 0 s

420 acked of 420 subscribed (cap 950)  
98 msg/s · last tick 0 s · delayed cluster  
NVDA 420/420 · focus none · rest 60 s

## Per-ticker sources

1 pinned · [clear](#)

A ticker pinned to another source fetches, streams and captures from it; the rest follow the universe source. Pick a source on the ticker's row, or add a name from another source's catalogue with the search's source selector.

NVDA → Massive

**24 September capture:** 420 acknowledged contracts against a configured 950-contract plan. The source is on a delayed cluster.

# Synchronising quotes to the current spot

An old option price should first be inverted using the forward associated with its own timestamp.

$$\Delta\sigma_{\text{naive}} \simeq -\frac{\Delta_F \Delta F}{\text{Vega}}$$

$$s_{\text{age}} = a\sqrt{\text{age}_{\text{min}}} \frac{\sigma_{\text{ATM}}}{0.15}$$

- **Price to variance:** Group observations by their quote spot, resolve the corresponding forward, remove exercise premium and invert.
- **Variance to the current frame:** Apply the chosen spot–vol rule using the last committed smile as a reference. Age adds an uncertainty allowance.

## ILLUSTRATIVE CALCULATION

At 15% ATM volatility and four minutes of age,  $a = 3.6$  vol bp per  $\sqrt{\text{minute}}$  gives a 7.2 vol bp age allowance.

### IN THE APP

Options ▶ Calibration: the Quote synchronisation switch and the age uncertainty in bp per  $\sqrt{\text{minute}}$ .

# Forward and discount from put–call parity

Call–put pairs at several strikes identify the intercept and slope of a line.

$$C(K) - P(K) = D(F - K) = a + bK$$

$$D = -b, \quad F = \frac{a}{D}$$

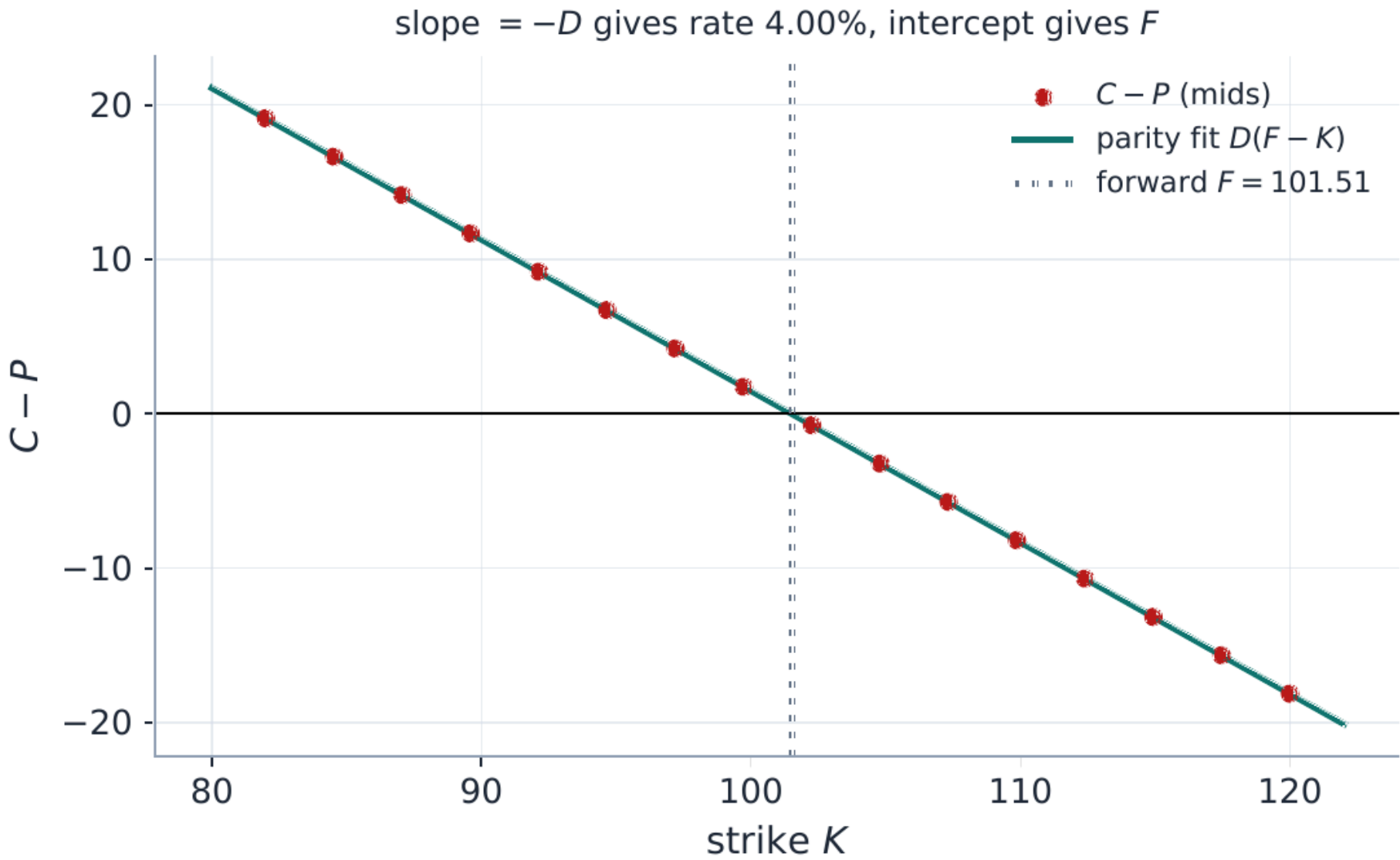
- **Observation set:** Use paired calls and puts from the same expiry and compatible option series.
- **Estimation:** Regress the call–put difference on strike, trim large residuals and inspect the remaining dispersion.
- **American options:** Early-exercise premia alter the line; a de-biasing step links the parity estimate to the exercise model.

ILLUSTRATIVE CALCULATION

If  $C - P = 101 - 0.98K$ , then  $D = 0.98$  and  $F = 103.0612$ . The zero crossing is the forward.

IN THE APP

Forwards lens (Alt+2): the ladder's Parity, Theo and Active columns and the residual count per expiry; the Forward panel's Parity / Theo / Manual mode.



**Reference-note figure:** the fitted line exposes the two quantities being inferred. Residuals measure inconsistency among the selected pairs.

# Dividends, borrow and the American fixed point

A parity forward becomes a carry estimate only after choosing the dividend and rate inputs.

$$F_{\text{theo}}(b) = (S - \text{PV}_{\text{div}})e^{(r-q-b)t}$$

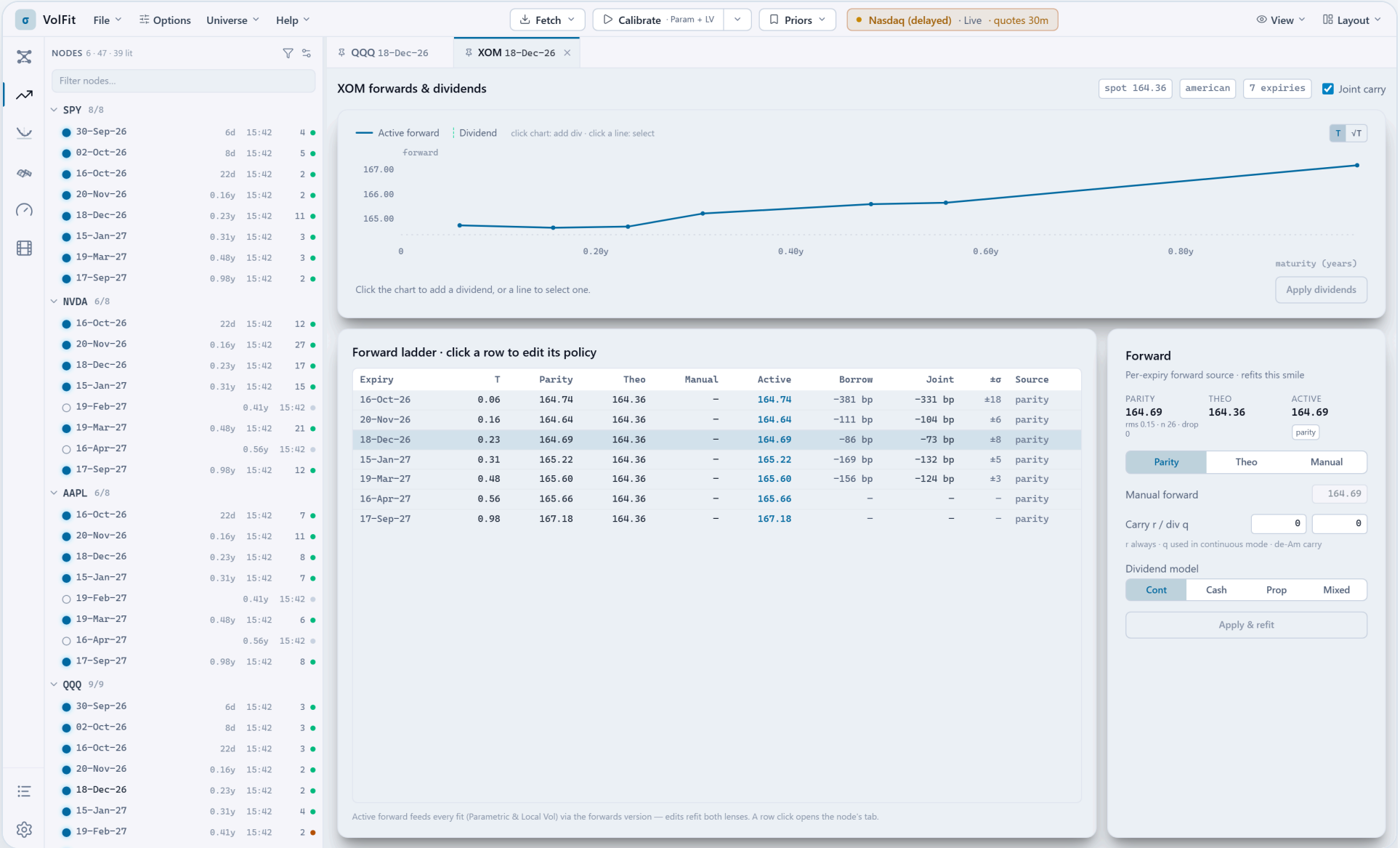
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$$b_{n+1} = b_n + \frac{1}{t} \log \left( \frac{F_{\text{theo}}(b_n)}{F_{\text{parity}}(b_n)} \right)$$

- **Dividend input:** Continuous yield, cash dates and amounts, proportional payments, or a mixture.
- **Coupled estimation:** Borrow changes the exercise calculation; the corrected option prices change the parity forward. Iterate until the two forwards agree.
- **Interpretation:** The resulting borrow includes errors in the supplied dividend and rate assumptions. Short maturities amplify forward noise.

IN THE APP

Forwards lens: tick Joint carry — the Joint and ±σ columns appear; hover a cell for the iterations and the ATM-vol sensitivity per 100 bp of borrow.



24 September XOM capture. The Joint and ±σ columns show the carry estimate and its reported uncertainty; the example uses the saved desk inputs.

# Removing the early-exercise premium

Estimate a European-equivalent volatility from the American mid, then remove the corresponding premium.

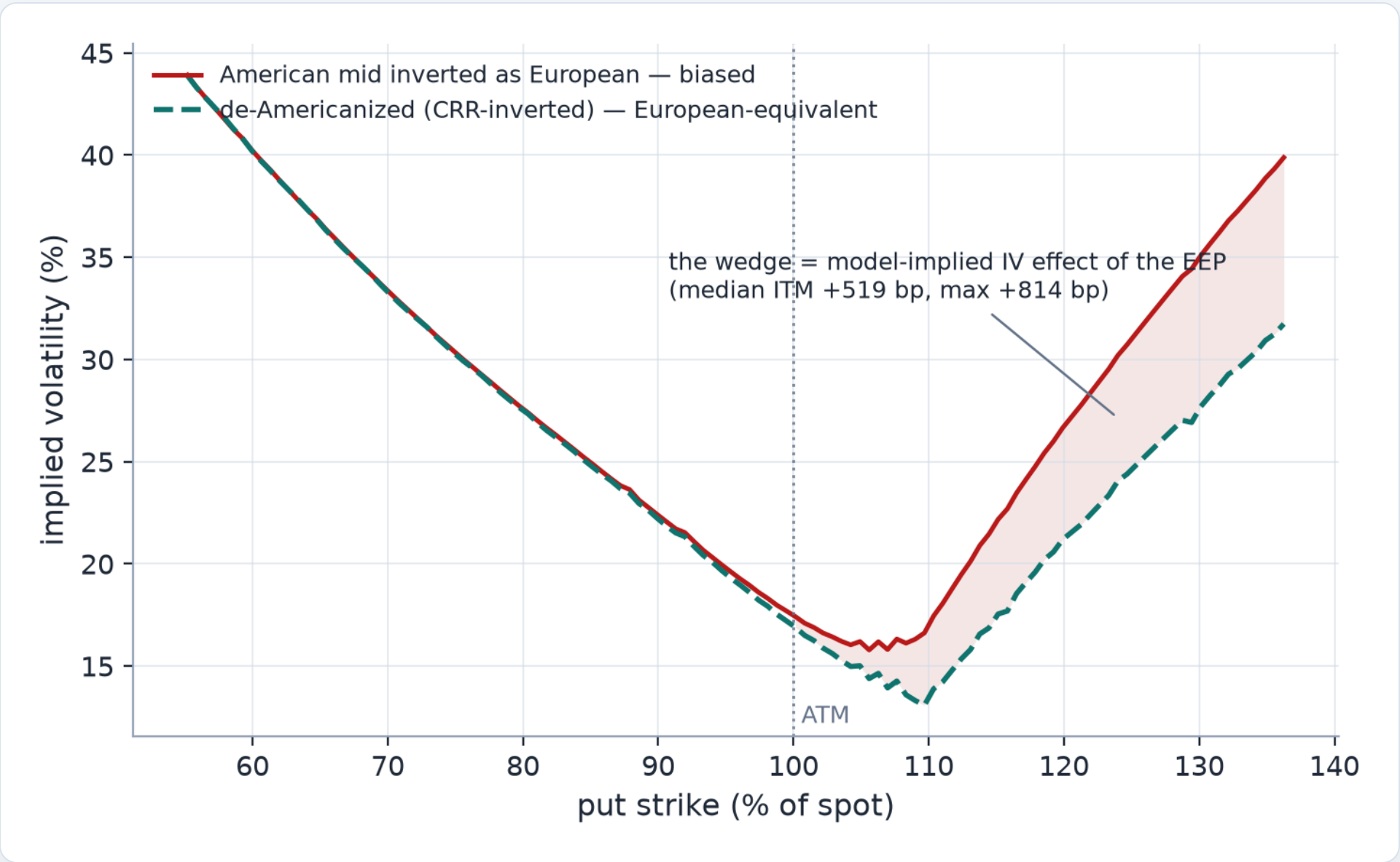
$$A_{\text{CRR}}(\sigma^*) = A^{\text{mid}}$$

$$\widehat{\text{EEP}} = \max(A^{\text{mid}} - E_{\text{Black}}(\sigma^*), 0)$$

- **One inversion per selected strike:** Fit the out-of-the-money side: puts below the forward and calls above it.
- **A common premium:** Subtract the estimated dollar premium from bid, mid and ask, then invert the European prices.
- **Exercise calculation:** The tree compares continuation with exercise at every step. Cash dividends enter through the escrowed-spot convention.

IN THE APP

Forward panel: switch the Dividend model to continuous with  $q = 0$  and Apply — call and put markers split at the money; restore the schedule and Calibrate.



**Recorded SPY example:** the largest displayed wedge is on deep ITM puts. The actual calibration population uses OTM quotes, where the correction is smaller.

# Exercise correction: numerical cost and accuracy

Root-solve precision, tree discretisation and quote uncertainty are different error sources.

| COMPONENT            | IMPLEMENTATION / RECORDED CHECK                    | INTERPRETATION                                      |
|----------------------|----------------------------------------------------|-----------------------------------------------------|
| American tree        | 192 CRR steps on the batch path                    | Fixed depth; $O(N^2)$ rollback per price            |
| Volatility root      | 24 bracketed bisections                            | About 0.002 vol bp bracket width at $\sigma \leq 4$ |
| Discretisation check | Within $\pm 5.4$ vol bp in the recorded round trip | Against American prices from a 2,001-step tree      |
| Batch execution      | Compiled kernel across quotes                      | About 0.07–0.10 ms per quote in the recorded runs   |
| Prepared-quote cache | Keyed on market data, carry and time inputs        | A change to fit settings can reuse preparation      |

RECORDED ACCURACY EXPERIMENT · 23 SEPTEMBER 2026

$S = 100$ ,  $r = 4.5\%$ ,  $q = 1.2\%$ ,  $t = 0.5$  year,  $\sigma = 25\%$ . The reference and batch paths use different tree depths; the comparison measures numerical resolution under that model.

# The units of a calibration residual

The objective measures how much the fitted model misses the selected market observations.

$$r_i(\theta) = \frac{c_\theta(k_i) - B(k_i, w_i)}{\partial_\sigma B(k_i, w_i) + \varepsilon}$$

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$$r_i \simeq \sigma_\theta(k_i) - \sigma_i$$

- **Price-based models:** LQD and local volatility price first. Vega normalisation expresses a small price error approximately in volatility units.
- **Volatility-based models:** SVI and MCS can form the volatility difference directly.
- **Low vega:** The denominator floor limits amplification. In that region, the residual behaves more like a scaled price error.

## ILLUSTRATIVE CALCULATION

A 0.04 price error with vega of 20 per unit of decimal volatility gives  $0.002 = 20$  vol bp. The approximation is local; a large miss also contains a vomma contribution.

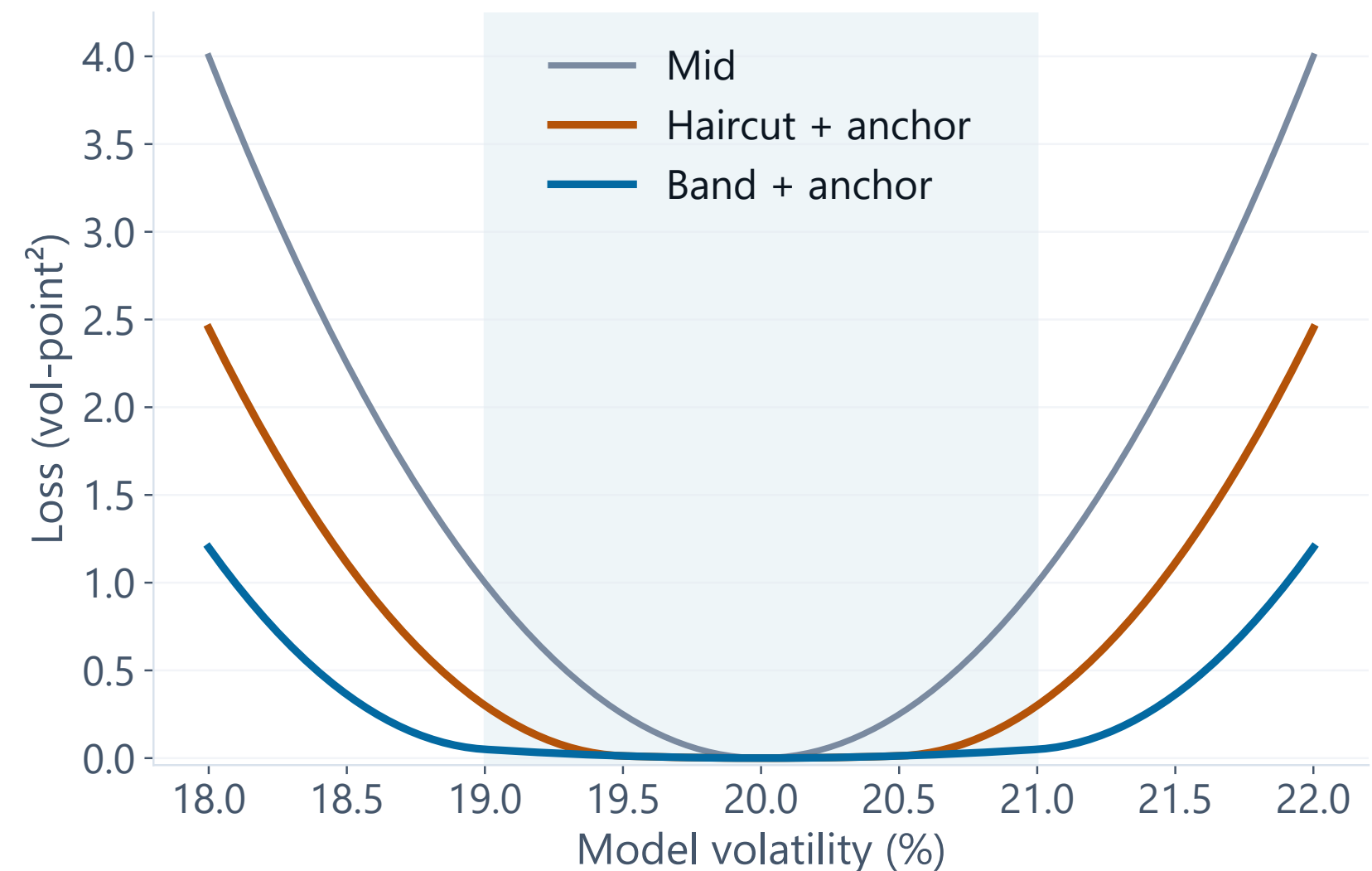
# Mid, bid–ask and haircut targets

A band target allows a range of fitted values, with a weak mid anchor selecting among them.

$$L_i = (m_i - u_i)_+^2 + (l_i - m_i)_+^2 + \theta_{\text{mid}}(m_i - \bar{m}_i)^2$$

- **Mid:** Minimise the squared difference to the quoted mid.
- **Band:** Penalise values outside the interval. Inside it, only the mid-anchor term remains.
- **Haircut:** Move each side toward mid by  $h$ , stopping at mid. A spread narrower than  $2h$  collapses to a mid target.

**IN THE APP** Show the target shading on the Smile, then the Mid / Bid–Ask / Haircut controls in Calibration options.



**Illustrative 19% / 21% quote.** A 0.5 vol-point haircut leaves [19.5%, 20.5%]. The band includes a 0.05 mid-anchor weight.

# Quote weights and strike spacing

A sum over listed strikes approximates a chosen distribution of weight along log-moneyness.

$$\omega_i \propto f(k_i) \min\left(\frac{s_i}{\bar{s}}, 10\right), \qquad \frac{1}{n} \sum_i \omega_i = 1$$

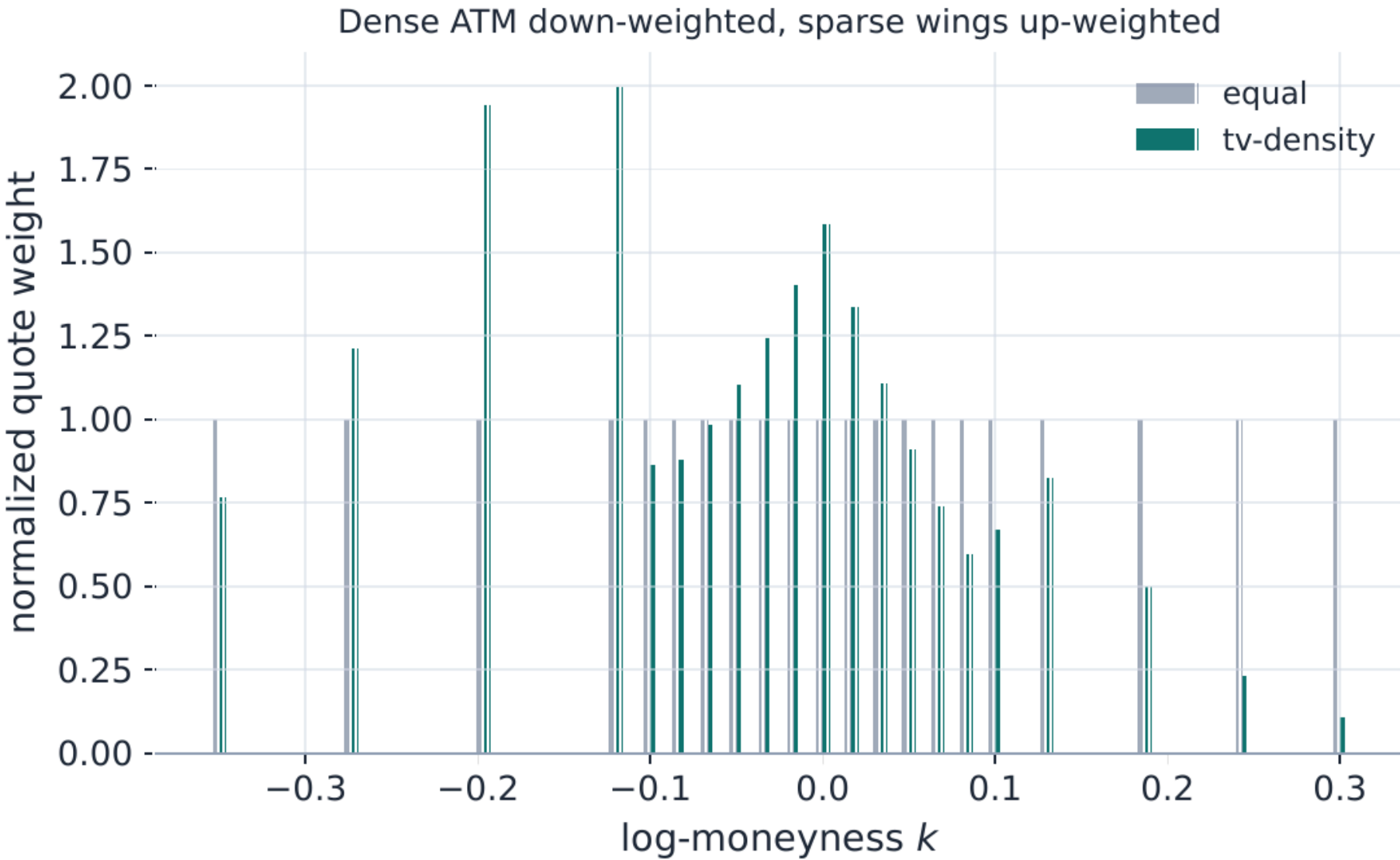
- **Target distribution**  $f$ : Uniform in log-strike, option time value, vega or OTM delta.
- **Spacing**  $s$ : A Voronoi cell measures the log-strike interval represented by one quote. Dense clusters receive smaller cells.
- **Equal weighting**: One unit per quote, without the spacing correction. This weights the actual listing density.

### ILLUSTRATIVE CALCULATION

Three quotes spanning 0.03 in  $k$  each represent roughly 0.01. A single quote representing 0.03 receives about three times an individual clustered quote's raw spacing weight, before caps and normalisation.

#### IN THE APP

Smile view: switch the Weights layer on in the layer rail — the strip under the chart shows the target beside the weight the fit sums; hover a bar for  $k \cdot \text{target} \cdot \text{spacing} \cdot \text{weight}$ .



Reference-note example of the spacing correction. A weighting scheme controls the allocation of fit error; the quote band controls tolerance.

# What each model parameterises

The four families place flexibility in different mathematical objects.

| MODEL            | UNKNOWN OBJECT                          | PRICING / INVERSION         | MAIN CONSTRAINTS OR CONTROLS                   |
|------------------|-----------------------------------------|-----------------------------|------------------------------------------------|
| LQD              | Log quantile density of log-return      | Integrate a probability law | Moment integrability; numerical pricing checks |
| SVI-JW           | Total implied variance $w(k)$           | Black formula               | Structural wing bounds; density checks         |
| MCS              | Variance profile with local corrections | Black formula               | Wing budget; positivity and density checks     |
| Local volatility | Local variance $\nu(\tau, x)$           | Forward Dupire PDE          | Positive vertices; roughness; lattice checks   |

## COMPARISON CONVENTION

Use the same prepared quotes, forward, variance clock, target mode and weighting. Record the tail and regularisation choices alongside the fit error.

# LQD: constructing a probability law

Let  $Q(u)$  be the quantile of  $X = \log(S_T/F_T)$ , with  $u$  between zero and one.

$$Q'(u) = q(u) = \frac{e^{g(u)}}{u(1-u)}$$

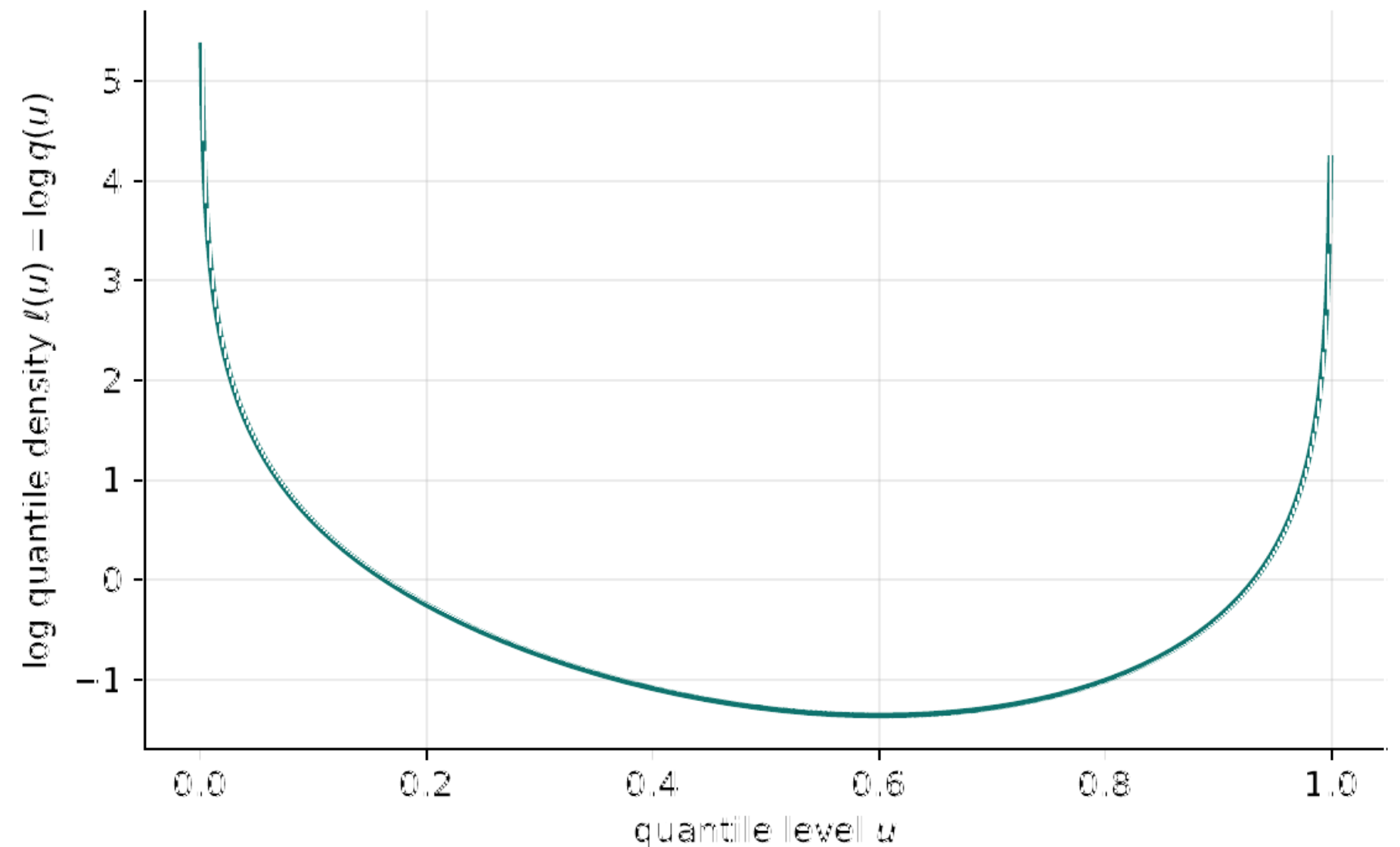
$$g(u) = (1-u)L + uR + \sum_{n=2}^N a_n P_n(1-2u)$$

$$f_X(Q(u)) = \frac{1}{q(u)} > 0$$

- **Construction:** Exponentiating  $g$  makes the quantile derivative positive. Integrating gives an increasing quantile.
- **Normalisation:** An additive shift  $\mu$  enforces  $\mathbb{E}[e^X] = 1$ . The exponential-tail case requires the right endpoint scale  $A_R < 1$ .

IN THE APP

Parametric ▶ Density ▶ Log Q-density: the two logarithmic walls at the ends and the Legendre body between them; Density and CDF are one click away.



**Reference-note figure:** the logarithmic endpoint terms set the tail structure; the smooth Legendre component controls the body.

# LQD: quantiles to option prices

A cumulative asset-share integral prices every strike after one construction of the slice.

$$z = \log \frac{u}{1-u}, \quad \frac{dQ}{dz} = e^{g(u(z))}$$

$$G(u) = \int_u^1 e^{Q(v)} dv$$

$$c(k) = G(u_k) - e^k(1 - u_k), \quad Q(u_k) = k$$

- **Numerical coordinate:** The log-odds coordinate  $z$  removes the endpoint singularities from the integration.
- **Price readout:** Locate the strike's rank  $u_k$ . Subtract the strike cash payment from the asset share above that rank.

## ILLUSTRATIVE CALCULATION

For  $g = \log a$ ,  $Q(u) = \mu + a \log[u/(1-u)]$ . If  $0 < a < 1$ ,  $\mu = -\log[\pi a / \sin(\pi a)]$ . At  $a = 0.10$ ,  $\mu \approx -0.01650$  and the mean of  $e^X$  is one.

# LQD calibration and analytic sensitivities

The strike-boundary term cancels when differentiating an option price.

$$\frac{dc}{d\theta} = \left. \frac{\partial G}{\partial \theta} \right|_{u_k}$$

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$$\left[ \frac{\partial G}{\partial u}(u_k) + e^k \right] \frac{du_k}{d\theta} = 0$$

- **Reason:** The option payoff is zero at its exercise boundary. Moving that boundary contributes no first-order price term.
- **Computation:** Cache ranks and basis values; integrate all parameter sensitivities in a batch. Optimise on 2,001 nodes and rebuild the accepted slice on 8,001.
- **Capacity:** The order guard reduces polynomial degree on thin slices. A degree-weighted ridge discourages high-order oscillation.

| WORK                    | SCALING / REUSE                                 |
|-------------------------|-------------------------------------------------|
| Slice and sensitivities | O(MP) for M grid nodes and P coefficients       |
| Many strikes            | Reuse the same cumulative arrays                |
| Nearby calibrations     | Warm starts reuse the previous parameter vector |

# ATM handles and delta-based operators

Local derivatives and option packages describe different aspects of the fitted smile.

$$h = (\sigma_0, s_0, \kappa_0) = (\sigma(0), \sigma'(0), \sigma''(0))$$

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$$RR_d = \sigma(k_{c,d}) - \sigma(k_{p,d})$$

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$$BF_d = \frac{\sigma(k_{c,d}) + \sigma(k_{p,d})}{2} - \sigma_0$$

IN THE APP

Options ▶ Prior: the Operators chips (ATM · RR25 · BF25 · RR10 · BF10 · WingL · WingR · VarSwap) and the Collar sign; the Fit diagnostics card reads the handles.

| QUANTITY                         | MEANING                                   | USE                         |
|----------------------------------|-------------------------------------------|-----------------------------|
| ATM / skew / curvature           | Level and local derivatives in k          | Filter and graph state      |
| RR25 / BF25; RR10 / BF10         | Differences across delta-selected strikes | Persistence operators       |
| Wing and variance-swap operators | Deep-strike slope or integrated smile     | Additional tail information |

ILLUSTRATIVE CALCULATION

ATM 20%, 25Δ call 19%, 25Δ put 23% gives  $RR_{25} = -4$  vol points and  $BF_{25} = +1$  vol point. At these fixed leg strikes, a common +2-point shift leaves RR and BF unchanged.

# SVI and Jump-Wings coordinates

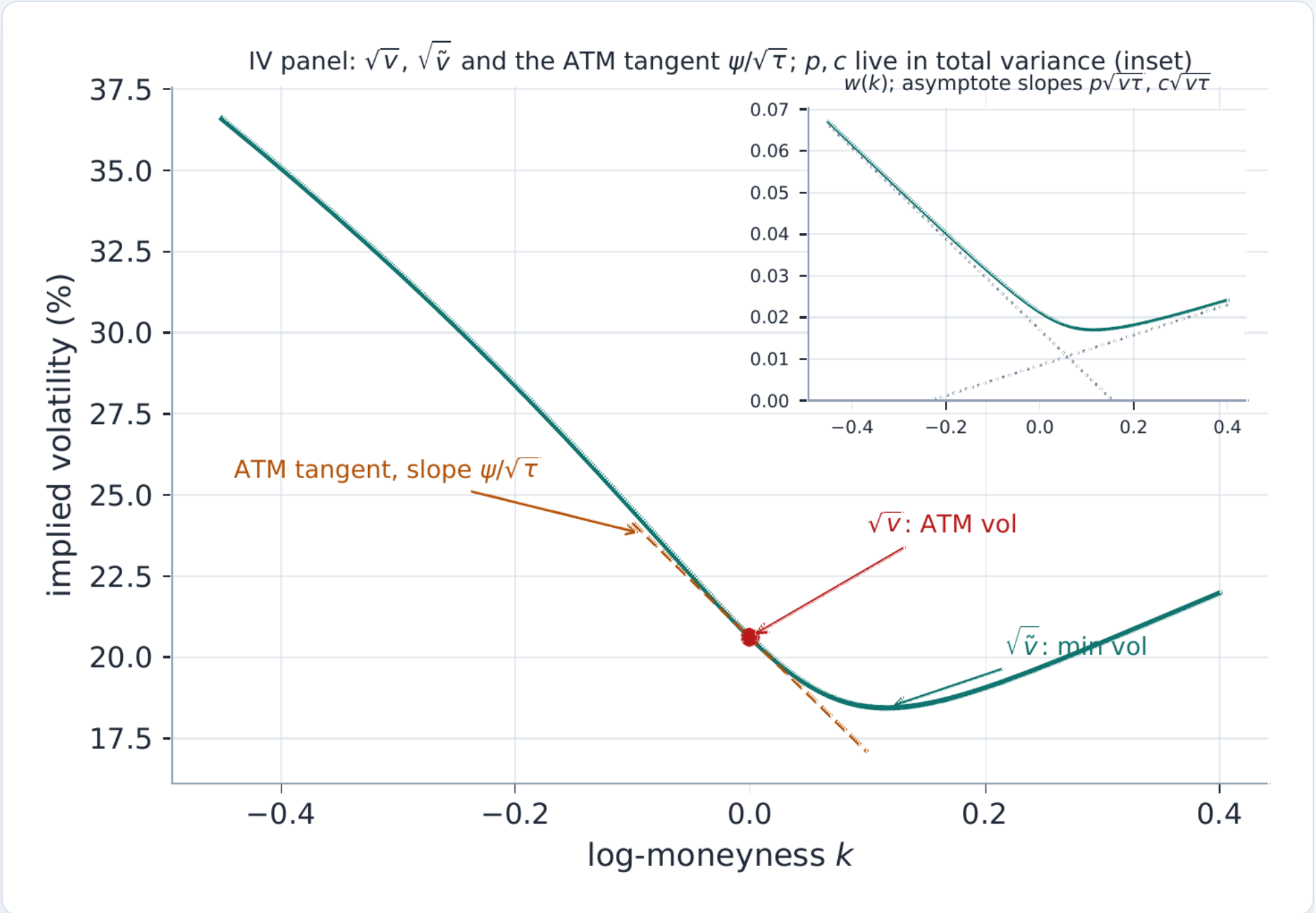
SVI fits a five-parameter total-variance curve; Jump-Wings expresses the same curve in more interpretable quantities.

$$w(k) = a + b \left[ \rho(k - m) + \sqrt{(k - m)^2 + \xi^2} \right]$$

- **Raw coordinates:**  $a$  sets the vertical offset,  $b$  the slope scale,  $\rho$  the asymmetry,  $m$  the centre and  $\xi$  the rounding scale.
- **Jump-Wings:** ATM variance, an ATM skew quantity, normalised left and right wing slopes, and minimum variance.
- **Admissibility:** The structural chart keeps the asymptotic slopes below the configured Lee cap. A separate density check examines the belly.

IN THE APP

Parametric ▶ Compare: click the SVI chip — the same quotes, one RMS column, Lee slopes and tails beside LQD; + reference adds eSSVI.



Reference-note curve annotated in Jump-Wings quantities: ATM variance and skew, the two wing slopes, and the minimum variance.

# MCS: local shape corrections

A smooth base is augmented with local corrections whose contribution vanishes in the far wings.

$$\Phi_{\kappa}(u) = \frac{4}{\kappa^2} \log \cosh\left(\frac{\kappa u}{2}\right)$$

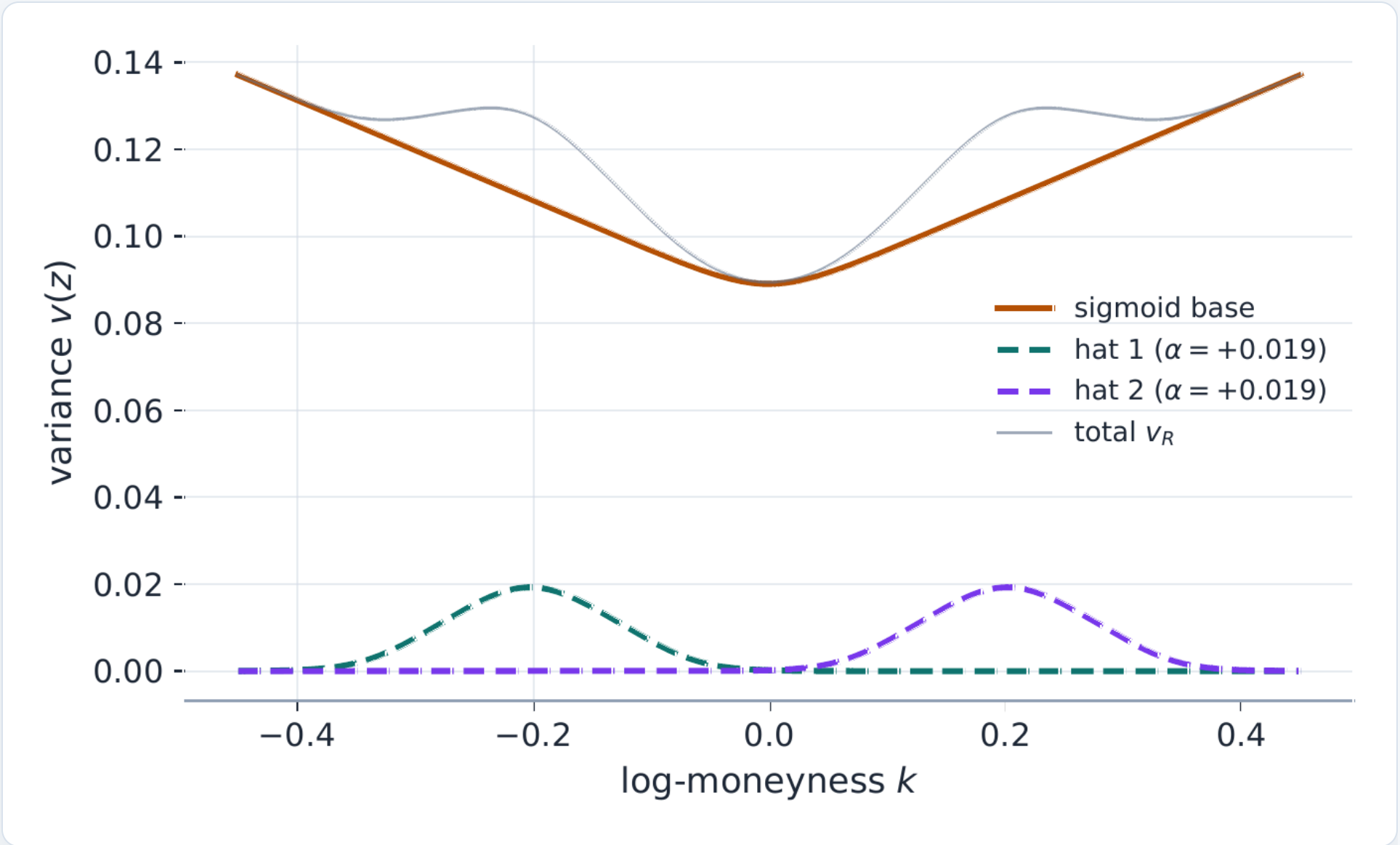
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$$H(u) = \frac{\Phi_{\kappa}(u - h) - 2\Phi_{\kappa}(u) + \Phi_{\kappa}(u + h)}{2\Phi_{\kappa}(h)}$$

- **Base:** A level, slope and log-cosh term provide the broad variance profile.
- **Correction:** The second difference forms a localized hat. Its coefficient, centre and width control an additional feature.
- **Fit controls:** Core count and ridge control flexibility. Density and wing conditions are evaluated on the resulting variance curve.

IN THE APP

Parametric ▶ Compare: the MCS chip; Options ▶ Parametric: MCS cores R and the wing penalty.



Reference-note decomposition. The hats alter the body while tending to zero as the base becomes linear in either tail.

# Generalised LQD tail exponents

A fixed exponent on each side changes the asymptotic return-tail class.

$$\frac{dQ}{dz} = e^{g(u)} \ell_-^{-\alpha_-} \ell_+^{-\alpha_+}, \quad 0 \leq \alpha_{\pm} \leq \frac{1}{2}$$

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$$\ell_- = 1 - \log u, \quad \ell_+ = 1 - \log(1 - u)$$

- **Policy:** Choose  $\alpha_-$  and  $\alpha_+$  per underlying, common across its expiries. Fit the remaining shape parameters within that chosen family.

IN THE APP

Options ▶ Parametric ▶ Tail  $\alpha_- \cdot \alpha_+$  with the presets Exp · Int · Gauss; the Quality node card reads Lee L/R and Tail order.

| EXPONENT ON ONE SIDE | LOG-RETURN TAIL         | FAR-WING TOTAL VARIANCE                   |
|----------------------|-------------------------|-------------------------------------------|
| $\alpha = 0$         | Exponential             | Asymptotically linear in $ k $            |
| $0 < \alpha < 1/2$   | Faster than exponential | Sublinear growth; limiting Lee slope zero |
| $\alpha = 1/2$       | Gaussian rate           | Tends to a constant set by the tail scale |

ILLUSTRATIVE CALCULATION

A tiny positive  $\alpha$  can resemble  $\alpha = 0$  over the quoted range while changing the limiting moment domain. Compare tail-policy scenarios at common quotes rather than estimating  $\alpha$  separately at every expiry.

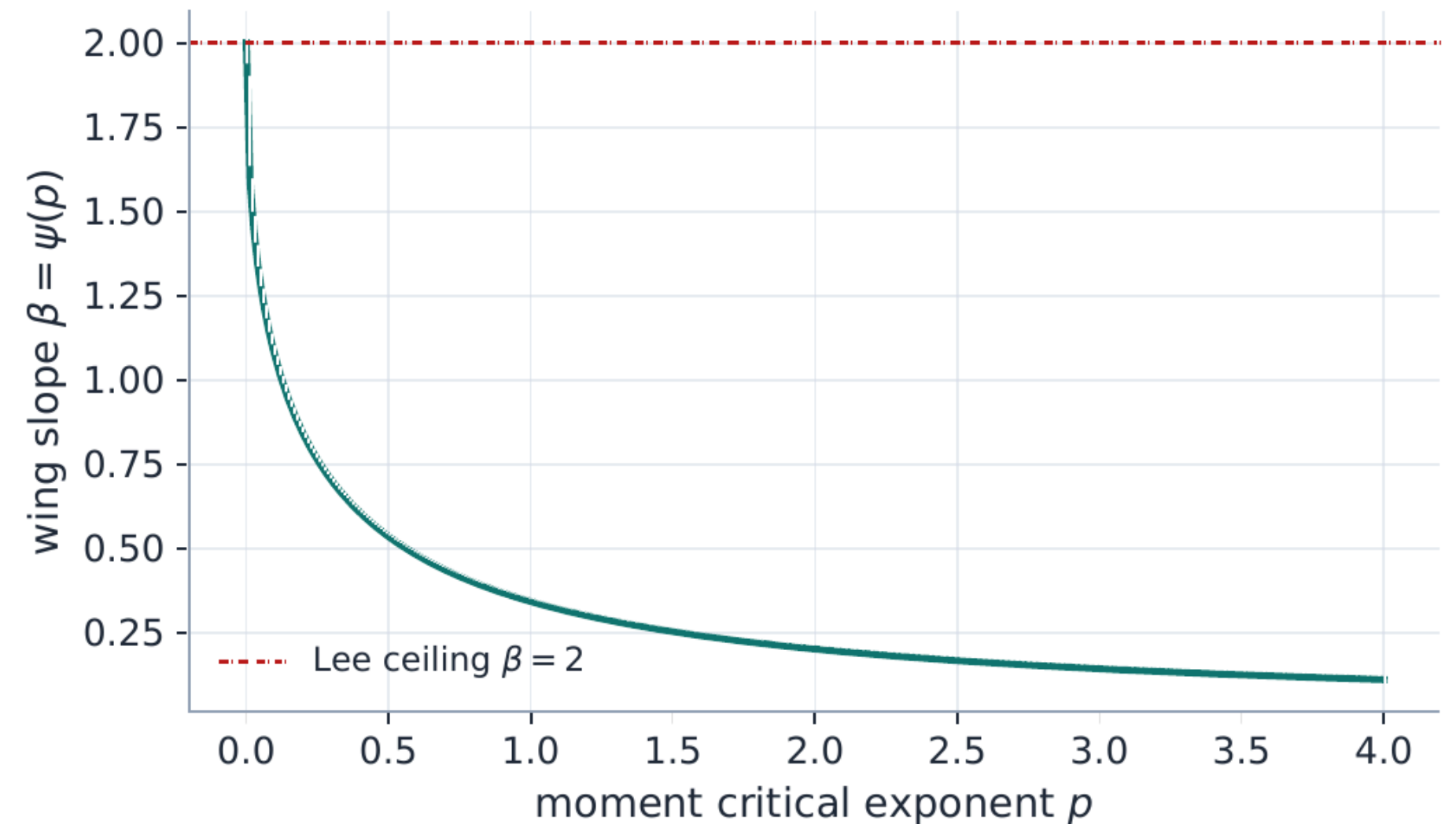
# Wing slopes and finite moments

Lee's relation connects asymptotic total-variance slopes to the moments of the normalized underlying.

$$\psi(p) = 2 - 4 \left( \sqrt{p^2 + p} - p \right)$$

$$\beta_R = \psi(p^*), \quad p^* = \sup\{p \geq 0 : \mathbb{E}[Y^{1+p}] < \infty\}$$

- **Left tail:** Use  $q^* = \sup\{q \geq 0 : \mathbb{E}[Y^{-q}] < \infty\}$  and  $\beta_L = \psi(q^*)$ .
- **Exponential LQD:**  $p^* = 1/A_R - 1$  and  $q^* = 1/A_L$ . Its tail scales therefore determine the limiting wing slopes.
- **Interpretation:** A limiting slope constrains remote tails. It does not identify the shape immediately beyond the final quote.



**Reference-note curve:** larger finite-moment orders correspond to smaller limiting wing slopes. The admissible bound is between zero and two.

# Variance swaps as an integrated constraint

A variance-swap quote adds information about the weighted option strip, including extrapolated wings.

$$w_{VS} = 2 \left[ \int_0^1 \frac{p(x)}{x^2} dx + \int_1^\infty \frac{c(x)}{x^2} dx \right]$$

$$\sigma_{VS} = \sqrt{w_{VS}/\tau}$$

$$r_{VS} = \sqrt{\lambda_{VS}}(\sigma_{VS}^{model} - \sigma_{VS}^{quote})$$

- **Meaning:** The log-contract identity links the full option strip to expected realised variance under continuous-path assumptions.
- **Calibration:** One residual compares the model's fair volatility level with the quoted level. Its weight is scaled to the node's total quote weight.
- **Tail allocation:** The fitted response depends on the model, the quoted strip and other penalties. One integrated quote does not uniquely determine both wings.

IN THE APP

Parametric aside ▶ Variance swap card: seed a quote at the model level, move it, watch the wing; Hard pin makes it a stiff row.

Redo

undo

edits

Variance swap

model 25.03% · quote ...

✕

A penalty pulls the fitted var-swap toward the quote.

penalty 15% of quote weight ≈ 33.8 abs · 1% of node RMS² · Options ▶ Calibration

replication strip 87% · tails L 13% / R 0%

Hard pin ☐

25.04

% vol

active

Exclude

Remove

Undo

Redo

Reset

24 September QQQ capture: quoted volatility 25.04%, model 25.03%, with a saved 15% penalty setting. These are recorded fit values.

# Calendar consistency across expiries

Individually valid slices also need a compatible order across maturity.

$$c_{T_2}(k) \geq c_{T_1}(k) \iff w_{T_2}(k) \geq w_{T_1}(k)$$

$$G_{T_2}(u) \geq G_{T_1}(u), \quad G_T(u) = \int_u^1 e^{Q_T(v)} dv$$

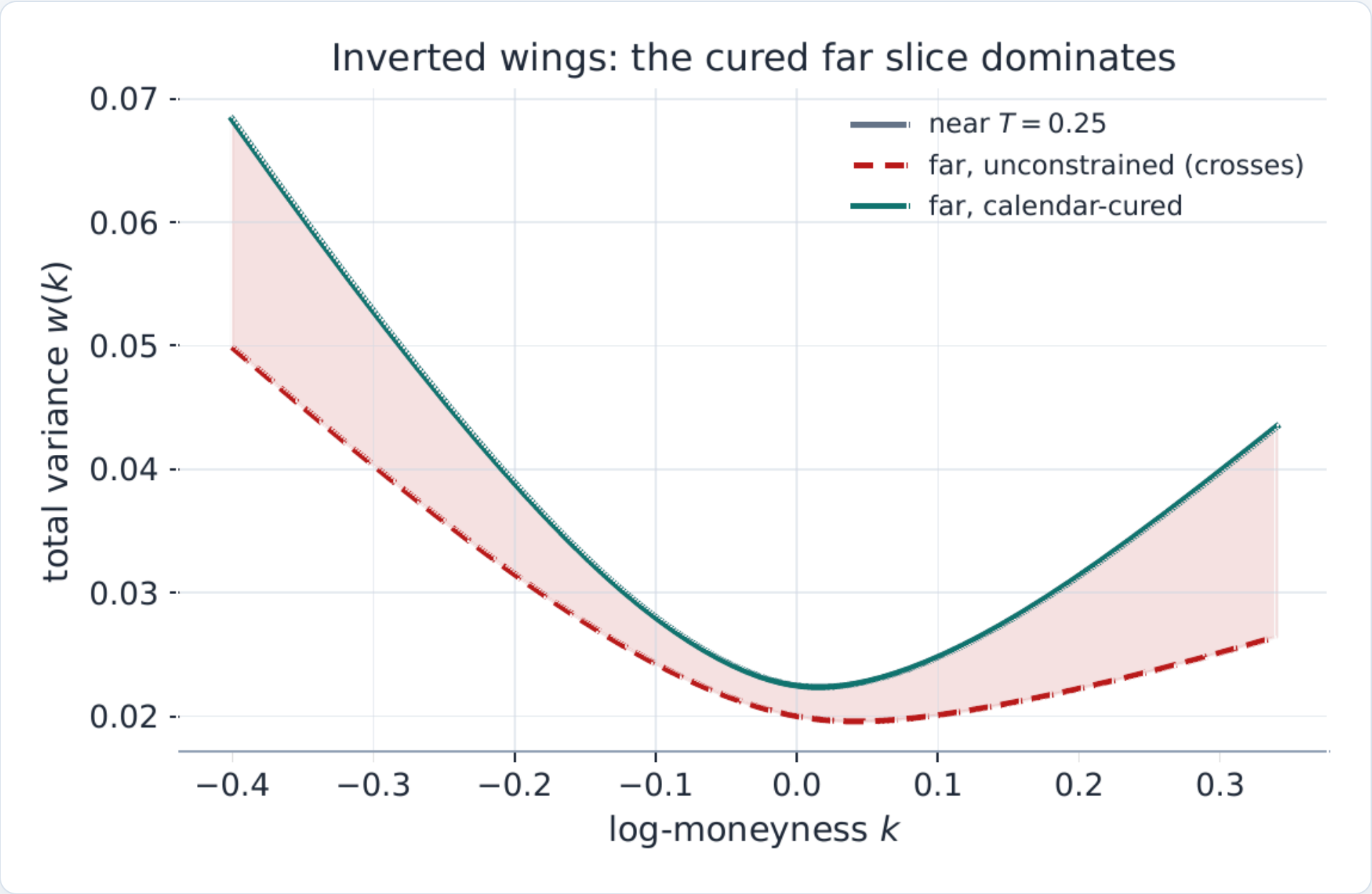
- **Mathematical condition:** Under the common mean-one martingale normalization, the farther distribution must dominate in convex order.
- **LQD diagnostic:** The integrated-quantile curve  $G$  is already available from pricing.
- **Volatility reading:** Compare total variance, not annualised volatility. A lower longer-dated volatility can still have greater total variance.

### ILLUSTRATIVE CALCULATION

Three months at 24% gives  $w = 0.0144$ . Six months at 20% gives  $w = 0.0200$ . The annualised volatility falls, while total variance rises.

IN THE APP

Parametric ▶ Stacked IV: total variance per expiry, non-crossing; Quality's Cal viol column and the calendar certificate per node.



Reference-note example of crossing expiry curves. A local negative difference in total variance identifies a calendar-order violation.

# Calendar repair and numerical certification

Fitting constraints and acceptance checks operate at different resolutions.

- **Certificate scope:** The implemented interior check isolates stationary points of the stored Hermite ledger gap. Tail order is reported separately and can be made a gate.

| STEP                      | CALCULATION                                          | PURPOSE                                           |
|---------------------------|------------------------------------------------------|---------------------------------------------------|
| Independent fits          | Fit each observed expiry with its own data terms     | Obtain a starting surface                         |
| Violation screen          | Compare adjacent expiries on relevant support        | Identify components needing joint repair          |
| Symmetric repair          | Optimize coupled expiries with calendar penalties    | Allow both sides to adjust within their fit costs |
| Stored-ledger certificate | Check gap extrema, boundaries and tail continuation  | Inspect between sampled grid points               |
| Policy decision           | Apply quote-region and full-line acceptance settings | State which violations block an exported surface  |

## ILLUSTRATIVE CALCULATION

For a cubic gap on one interpolation interval, minima occur at its endpoints or roots of its quadratic derivative. Checking only the stored nodes can miss an interior dip.

# A piecewise-affine local-variance surface

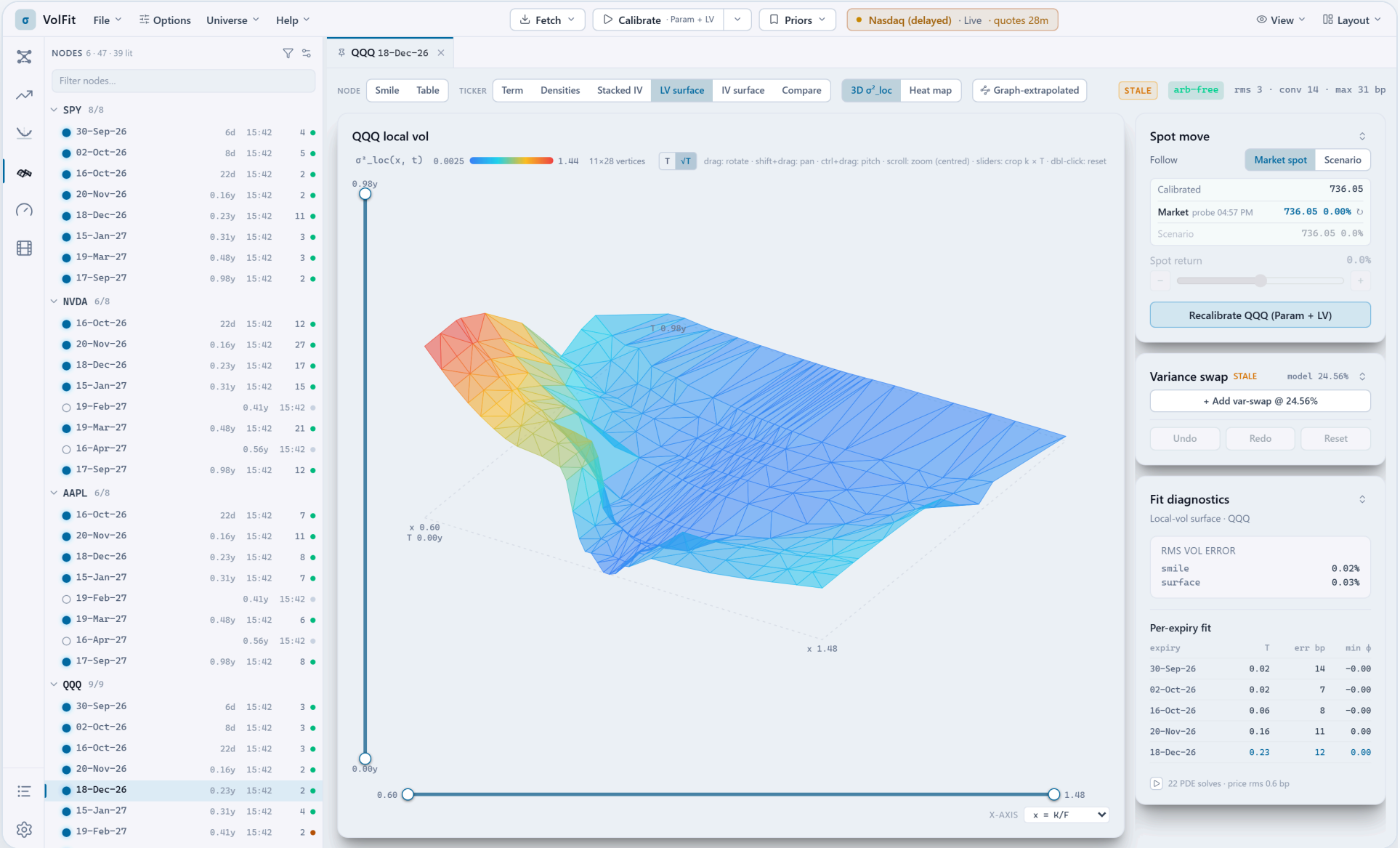
The local-volatility fit chooses instantaneous variances whose forward prices reproduce the option quotes.

$$\nu_{\theta}(\tau, x) = \sum_{\ell} \theta_{\ell} \phi_{\ell}(\tau, x), \qquad x = K/F$$

- **Parameters:** The  $\theta_{\ell}$  are local variances at the vertices of a triangulated strike–time grid.
- **Interpolation:** Within a triangle, barycentric weights are nonnegative and sum to one. Positive vertex values give positive interpolated variance.
- **Outside the grid:** A separate continuation rule specifies the wings. The right side is clamped; the left uses a configured linear extension.

IN THE APP

Local Vol lens (Alt+4) ▶ LV surface: drag to rotate the mesh; the toolbar badges arb-free and rms · conv · max bp.



24 September capture of the fitted local-variance mesh. The visible stale label refers to the saved snapshot; the mesh illustrates the parameterized object.

# The forward Dupire pricing calculation

Given local variance, march normalized call prices from the intrinsic payoff.

$$\partial_\tau c = \frac{1}{2} \nu(\tau, x) x^2 \partial_{xx} c, \quad c(0, x) = (1 - x)^+$$

---

$$(I - \gamma \Delta \tau A^{n+1}) U^{n+1} = a U^n - b U^{n-1}$$

- **Strike lattice:** Grade cells near the short-expiry density and include  $x = 1$  explicitly.
- **Time grid:** Use small initial steps near the payoff kink and include expiry and variance-vertex times.
- **Time stepping:** The recorded default is variable-step BDF2 with implicit-Euler startup and restarts. Each step uses a tridiagonal solve.

### ILLUSTRATIVE CALCULATION

At a constant BDF2 step size:  $\gamma = 2/3, a = 4/3, b = 1/3$ . The negative coefficient on  $U^{n-1}$  means an M-matrix on the left alone does not prove unconditional positivity of the full two-step update.

# Calibrating through the PDE

Forward sensitivities reuse each time-step factorization for all parameter columns.

$$MS_\ell^{n+1} = \frac{\partial \text{RHS}}{\partial \theta_\ell} - \frac{\partial M}{\partial \theta_\ell} U^{n+1}$$

$$L(\theta) = L_{\text{quotes}} + \lambda_x \|D_x^2 \theta\|^2 + \lambda_\tau \|D_\tau^2 \theta\|^2 + \lambda_d \|D_x^3 c\|^2$$

- **Derivative calculation:** Differentiate the discrete pricing equation. The same tridiagonal matrix  $M$  multiplies every sensitivity  $S_\ell$ .
- **Optimization:** Projected Gauss–Newton / Levenberg–Marquardt respects vertex bounds; active band rows are re-evaluated during a step.
- **Regularisation:** Spacing-aware variance roughness and call-price density-smoothness rows select among weakly identified sheets.

## COMPUTATIONAL INTERPRETATION

With  $M_x$  strike nodes,  $M_\tau$  time steps and  $P$  active variance parameters, a full forward sensitivity march costs roughly  $O(M_x M_\tau P)$ . Shared factorization reduces constants, not the parameter dimension.

IN THE APP

Local Vol aside ▶ Fit diagnostics: the per-expiry error table, N PDE solves · price rms, and the ▶ calibration-trace player.

# Fitted local variance and the Dupire-derived surface

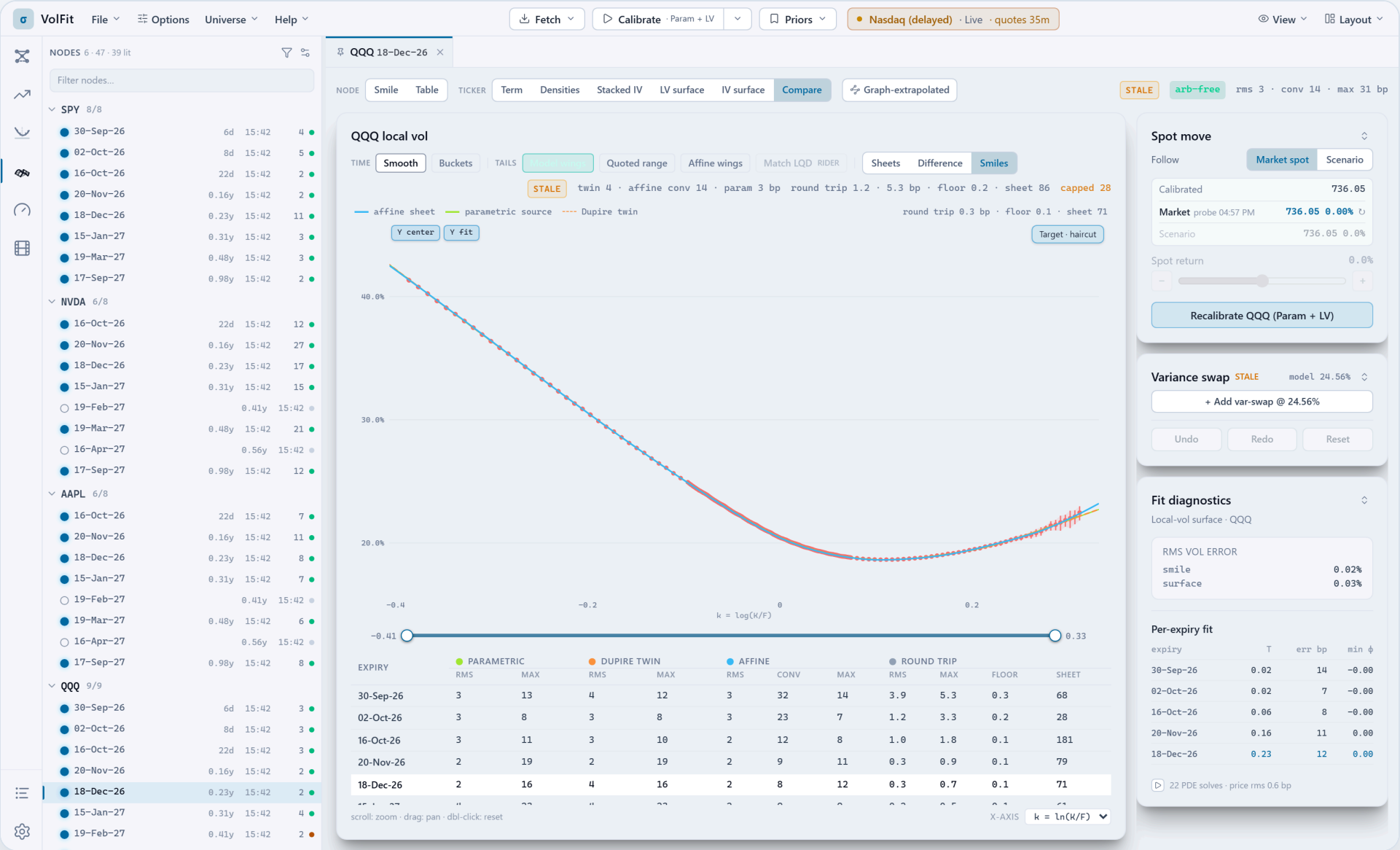
The Compare view separates the inverse fit from differentiation of a parametric surface.

- **Direct fit:** Choose local variance and price through the PDE to match the quotes.
- **Dupire-derived surface:** Differentiate the parametric call surface, then reprice it through the PDE to measure the round trip.
- **Refined reprice:** Evaluate the fitted local-variance sheet on a finer pricing grid. The change in error estimates operator sensitivity.

RECORDED SPY MEASUREMENT · 23 SEPTEMBER 2026

Nine expiries, 875 quotes, 253 vertices: 3.61 vol bp RMS on the fit operator and 5.12 bp on the refined reprice. The second number reveals a discretisation contribution.

IN THE APP Local Vol ▶ Compare ▶ Smiles: the three curves per expiry and the score table Parametric · Dupire twin · Affine · Round trip.



24 September QQQ Compare capture: the source parametric smile, its Dupire-derived reprice, and the directly fitted local-volatility smile.

# Event-weighted variance time

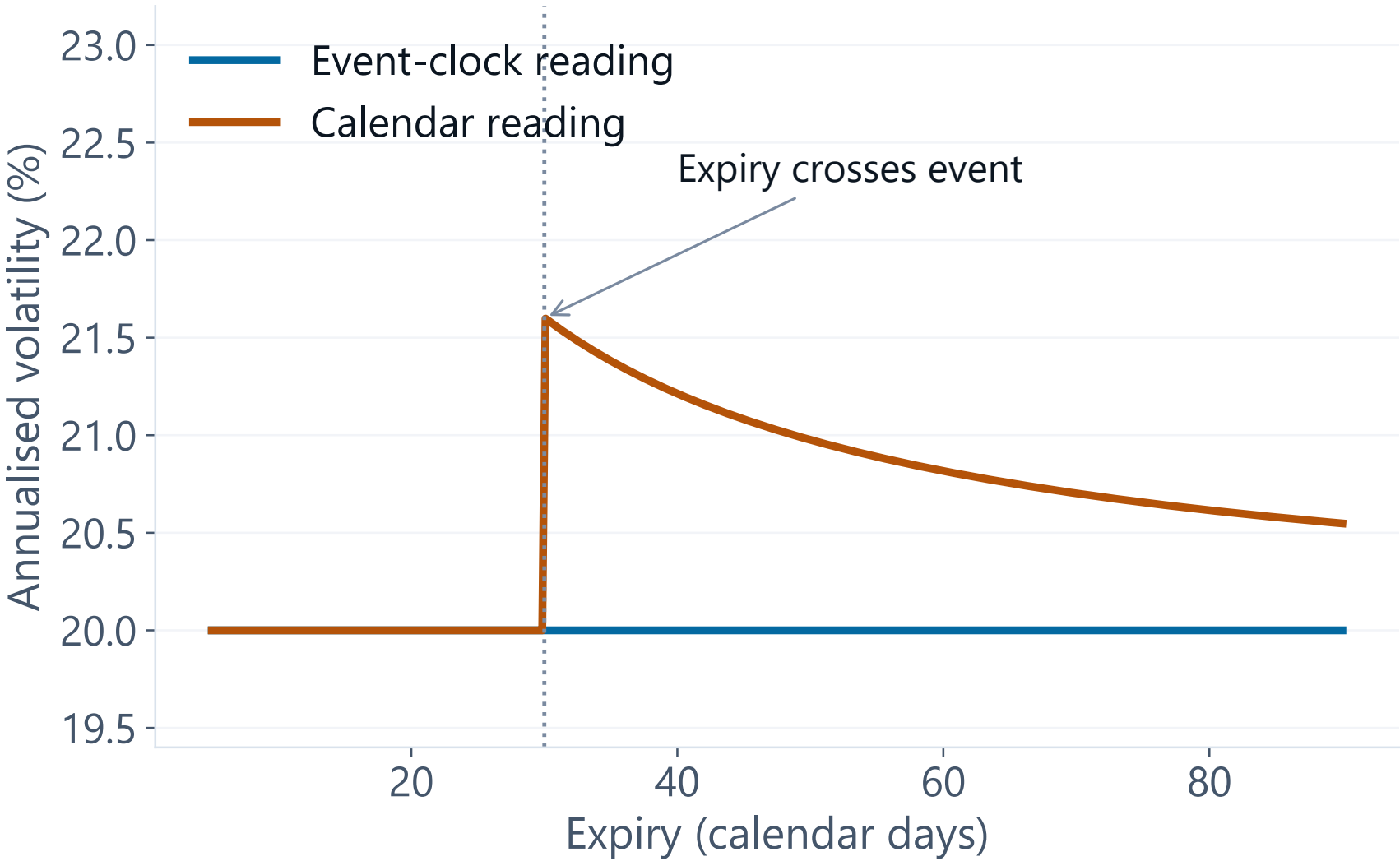
The event clock changes the time denominator used to report a given total variance.

$$\tau_{\text{raw}}(t) = t + \frac{1}{365} \sum_{t_e \leq t} N_e$$

$$\sigma_{\text{calendar}} = \sqrt{w/t}, \quad \sigma_{\text{event}} = \sqrt{w/\tau}$$

- **Event weight:**  $N_e$  adds equivalent variance days when expiry lies after the event.
- **Optional normalisation:** Divide by the one-year clock budget so the normalized clock has  $\tau(1 \text{ year}) = 1$ .
- **Two roles:** A clock toggle re-expresses a fixed price. A refit under a different clock can change the calibrated result through its objective and constraints.

IN THE APP

 Use the Term view's two clock readings and inspect the event date and extra-day weight.

**Illustrative 30-day expiry with five extra event days:** a 20% event-clock volatility corresponds to 21.60% on the calendar clock, without budget normalisation.

# Auto-calibrating isolated variance peaks

The expiry ladder identifies excess variance in an interval, rather than the exact event date.

$$f_i = \frac{w_i - w_{i-1}}{d_i}, \quad N_i = d_i \left( \frac{f_i}{r_i} - 1 \right)$$

- **Detection:** Compare each interval's variance per calendar day with its neighbouring intervals. An isolated local peak is a candidate.
- **Sizing:** Use the hotter neighbour as reference  $r_i$ . Add enough equivalent days to lower the interval rate to that reference.
- **Placement:** Put the event at the interval midpoint, then allow the date to be edited from external knowledge.

IN THE APP

Parametric ▶ Term: Auto-calibrate events with its horizon — the result line, the Ladder spread readout, and both clock readings on the chart. Use an all-lit ladder.

| ILLUSTRATIVE INTERVAL | DAYS $d$ | VARIANCE PER DAY $f$ | EXTRA DAYS |
|-----------------------|----------|----------------------|------------|
| Before                | 30       | 0.0001               | 0          |
| Peak                  | 30       | 0.0002               | 30         |
| After                 | 30       | 0.0001               | 0          |

# Spot transport between calibrations

The SSR rule translates the stored total-variance curve in current log-moneyness.

$$w_1(k) = w_0(k + Rh), \quad h = \log(F_1/F_0)$$

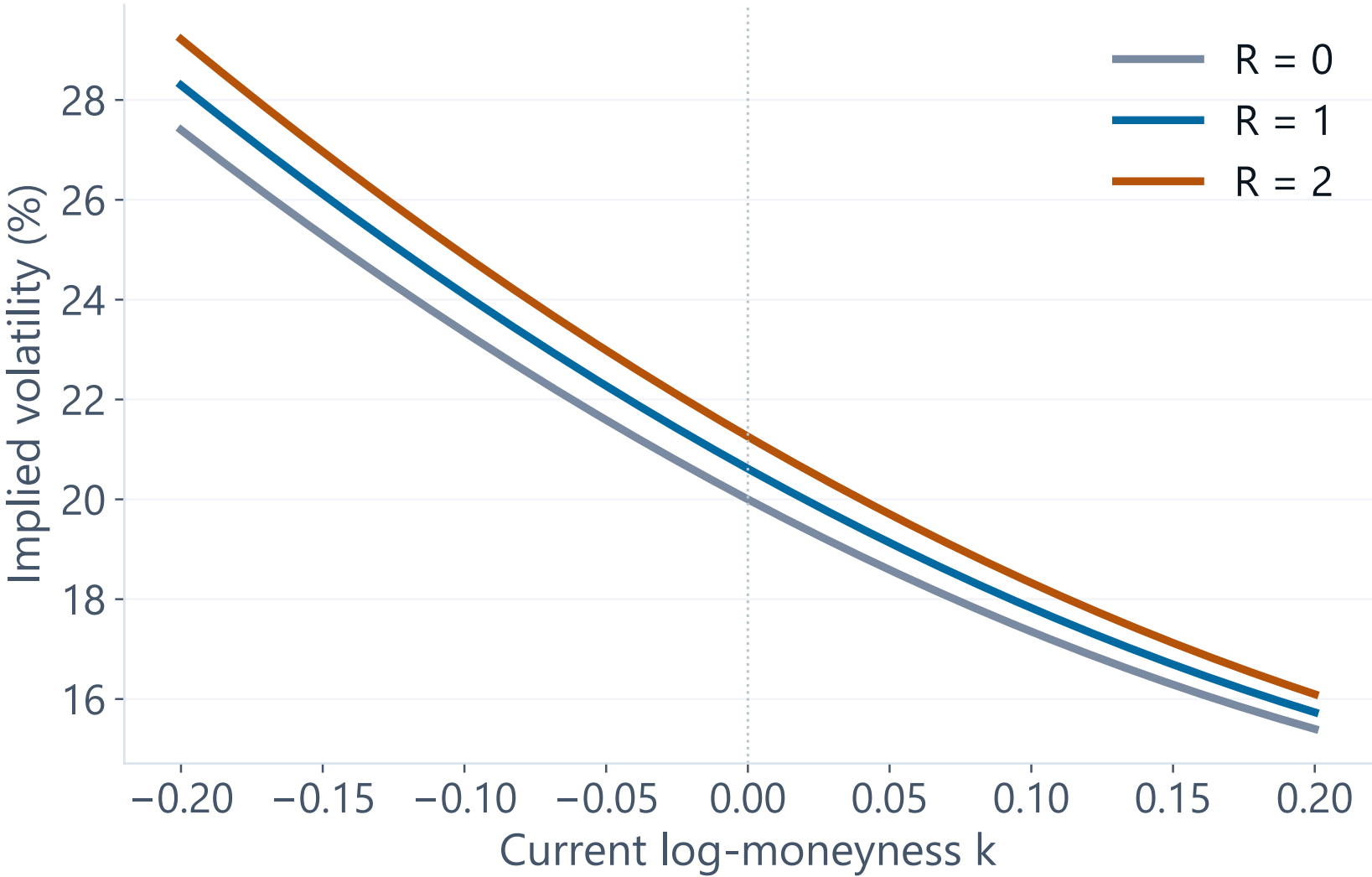
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$$\Delta\sigma_{\text{ATM}} \simeq R s_0 h$$

- $R = 0$ : Sticky moneyness: the curve stays fixed as a function of  $k$ .
- $R = 1$ : Sticky strike: volatility stays fixed at a given cash strike, for an unchanged clock.
- **Local-volatility choices:** The LV approximation uses a nonlinear strike map. The grid regime reprices a frozen local-volatility grid through the PDE.

IN THE APP

Spot move card ▶ Scenario: the  $\pm 0.1\%$  dial transports every lens; Options ▶ Dynamics switches the regime Mny · Strike · LV · LV grid · SSR while the fit stays.



Illustrative  $h = -2\%$ , ATM skew  $s_0 = -0.30$ . First-order ATM changes are 0, +60 and +120 vol bp for  $R = 0, 1$  and  $2$ .

# Activating a prior where quote support is weak

Each persisted operator receives a gate based on the support around the strikes it uses.

$$g_j = \left[ \text{clip} \left( 1 - \frac{\pi_j^{\text{obs}}}{\pi^{\text{req}}}, 0, 1 \right) \right]^\gamma$$

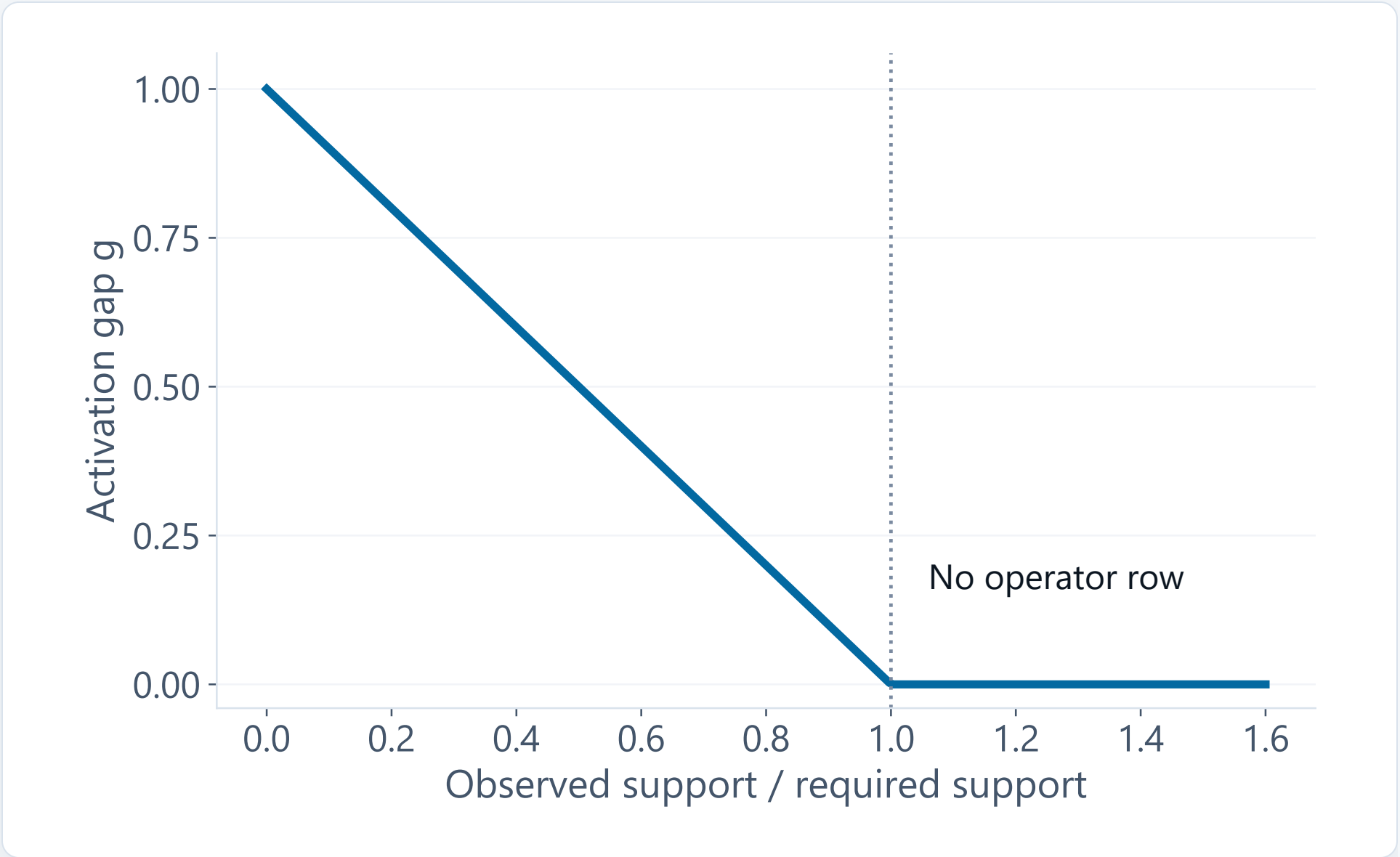
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$$r_j = \sqrt{\lambda_j} \frac{O_j(\theta) - O_j(\text{prior})}{s_j}, \quad \lambda_j = B \frac{g_j}{\sum_m g_m}$$

- **First transport:** Evaluate the saved prior at the current forward under the chosen dynamics rule.
- **Then gate:** An operator whose support exceeds the threshold receives no persistence row. Open gates share a configured budget.
- **Scope:** Quote support is a proxy for identification. It is not the exact null space of the fit's Hessian.

IN THE APP

Options ▶ Prior ▶ Evidence: per expiry and operator, gap · λ · age · src · h — gap near zero where the quotes identify the operator.



Illustrative  $\gamma = 1$  gate. Closing one operator's gate removes that residual; parameter coupling can still transmit effects from other active rows.

# Persisting shape versus absolute strike values

The choice of operator determines which market moves the prior resists.

$$RR(\sigma + c) = RR(\sigma), \quad BF(\sigma + c) = BF(\sigma)$$

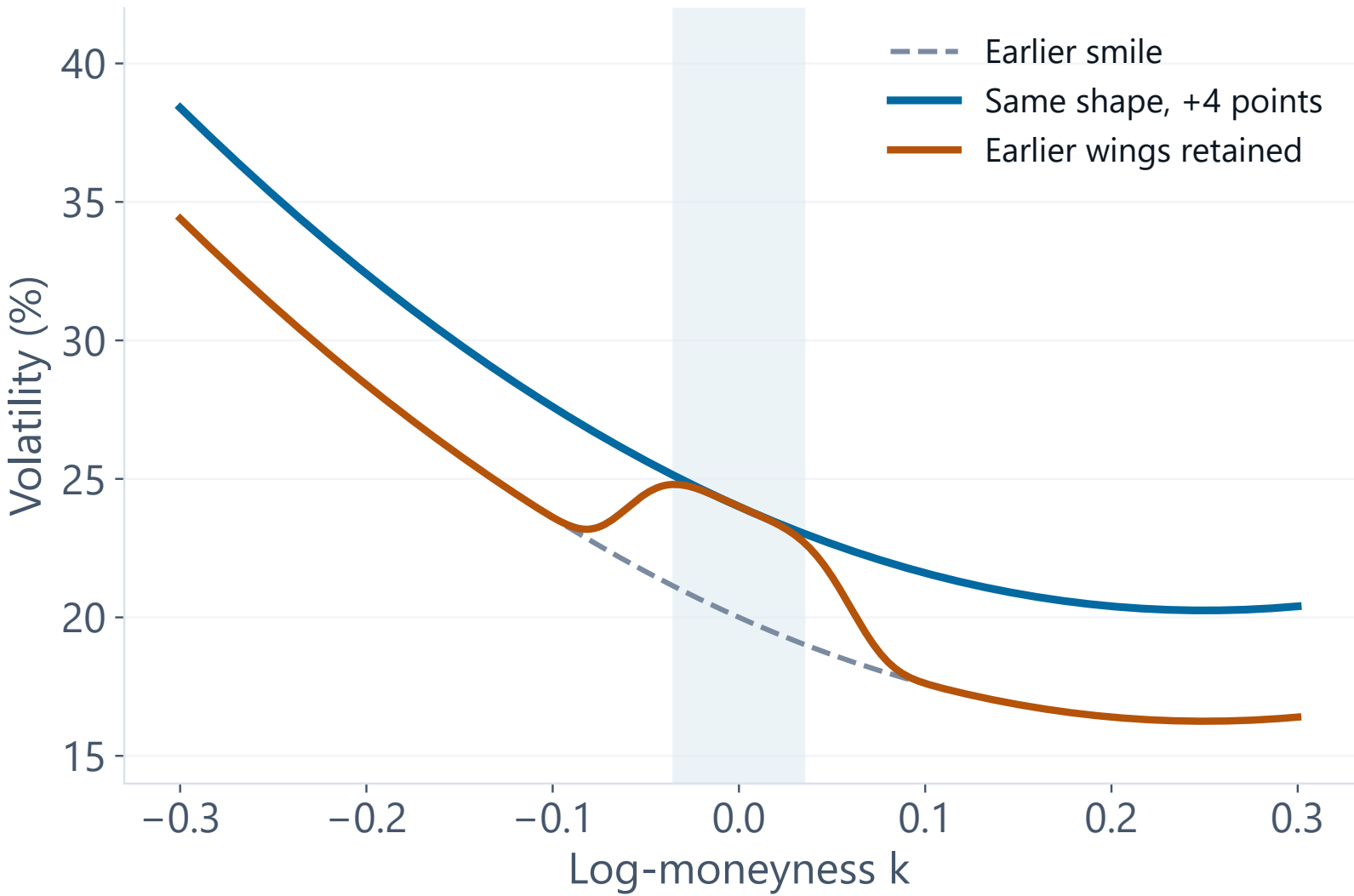
- **Shape baskets:** At frozen leg strikes, RR and BF leave a common volatility level shift unchanged.
- **Strike anchors:** A residual to yesterday's price or volatility at an unquoted strike resists a level shift there.
- **Hybrid:** Combine operator persistence with deeper strike anchors. Their roles and possible interactions remain distinct.

### ILLUSTRATIVE CALCULATION

Yesterday's 25Δ call / put = 19% / 23%. After a common +4-point move they are 23% / 27%; RR remains −4 points. An absolute 23% put anchor would oppose the new 27% level.

IN THE APP

On a thin node, compare the Free and +Prior views, then inspect which operator and strike-anchor rows are active.



**Illustrative curves:** at fixed leg strikes, the +4-point shift preserves RR and BF. An absolute wing anchor retains the earlier level outside the shaded quote region.

# Temporal filtering of ATM handles

The filter combines a transported prediction with a new estimate of level, skew and curvature.

$$K_t = P_t^- (P_t^- + R_t)^{-1}$$

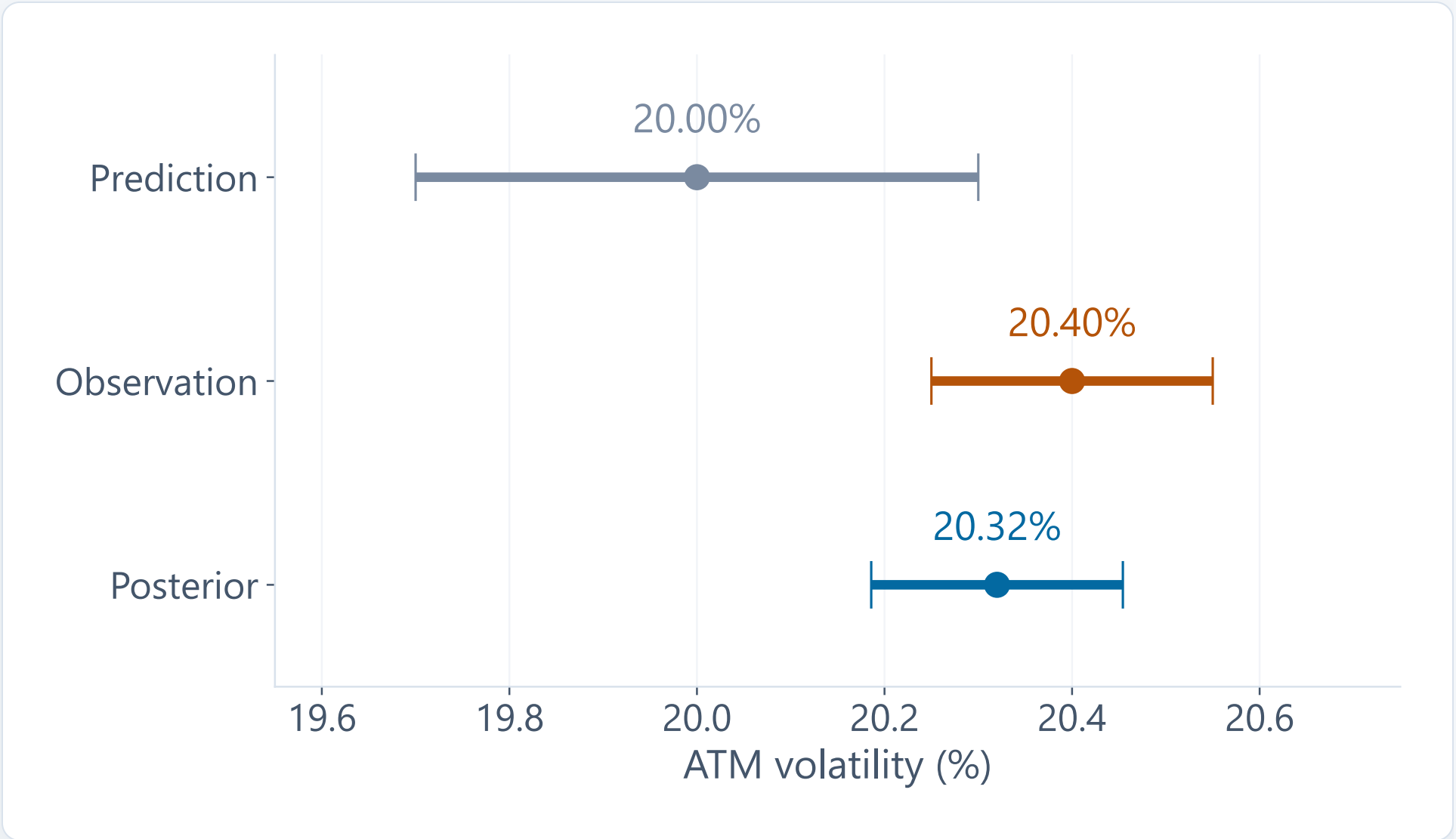
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$$m_t^+ = m_t^- + K_t(z_t - m_t^-)$$

- **Prediction uncertainty**  $P^-$ : Accumulates process noise with elapsed time, spot transport and configured surprise adjustments.
- **Observation uncertainty**  $R$ : Reflects the fitted quotes' stated noise and the sensitivity of the handle estimates.
- **Output**: A posterior handle vector, covariance and innovation diagnostic. Gains can differ strongly between level and curvature.

IN THE APP

Options ▶ Kalman filter: Overlay only, the diagnostics table  $K(\text{ATM}) \cdot K(\text{skew}) \cdot K(\text{curv}) \cdot \text{innov bp} \cdot \rho$ , and the Timeline; the smile's FILTER badge.



**Illustrative ATM update:** prediction 20.0% ± 0.30%, observation 20.4% ± 0.15% (1σ).  $K = 0.80$  and the posterior mean is 20.32%.

# Observation covariance and active filtering

Noise assumptions enter before the gain is computed; using the filtered state in calibration changes the objective.

$$R_x \simeq \rho G \mathcal{I}_\theta^{-1} G^\top, \quad G = \frac{\partial h}{\partial \theta}$$

---

$$L_{\text{active}} = L_{\text{quotes}} + \frac{1}{2} \|h(\theta) - m^-\|_{(P^-)^{-1}}^2 + L_{\text{other}}$$

| MODE / INGREDIENT       | ROLE                                                                            |
|-------------------------|---------------------------------------------------------------------------------|
| Information matrix      | Quote Jacobian scaled by stated quote noise, plus intrinsic rows                |
| Regularised inverse     | Clamp small eigenvalues before inversion; retain uncertainty in weak directions |
| Misfit inflation $\rho$ | Widen observation covariance when quotes disagree with their stated noise       |
| Use modes               | Overlay displays the update; active MAP uses the prediction once inside the fit |

RECORDED OPERATING LIMITATION

The September Series experiments found severe slowdowns and fit degradation for active filtering on short expiries at intraday cadence. Overlay mode permits inspection without that feedback.

# Series replay and independent calibration lanes

A saved sequence lets different model and memory settings consume the same observations in time order.

| OBJECT       | STORED / CARRIED INFORMATION                         | COMPARISON PURPOSE                         |
|--------------|------------------------------------------------------|--------------------------------------------|
| Frame        | Quotes, spot, as-of time and source                  | Hold the market inputs fixed               |
| Lane         | Model and calibration settings                       | Change one policy or model choice          |
| Lane history | Its own prior and filter state                       | Prevent one variant feeding another        |
| Replay       | Chronological calibration and transport              | Measure response, fit error and runtime    |
| Readouts     | Smiles, term structure, handles and lane differences | Locate when and where the variants diverge |

## ILLUSTRATIVE CALCULATION

Run Free, +Prior and +Prior with filter overlay on identical frames. Compare quote error and frame-to-frame handle movement separately; a smoother path alone is not proof of a better estimate.

IN THE APP

Series lens (Alt+6): New series... with the lane presets, then Lanes — the metric chart and the summary table across frames.

## Graph state: changes from transported priors

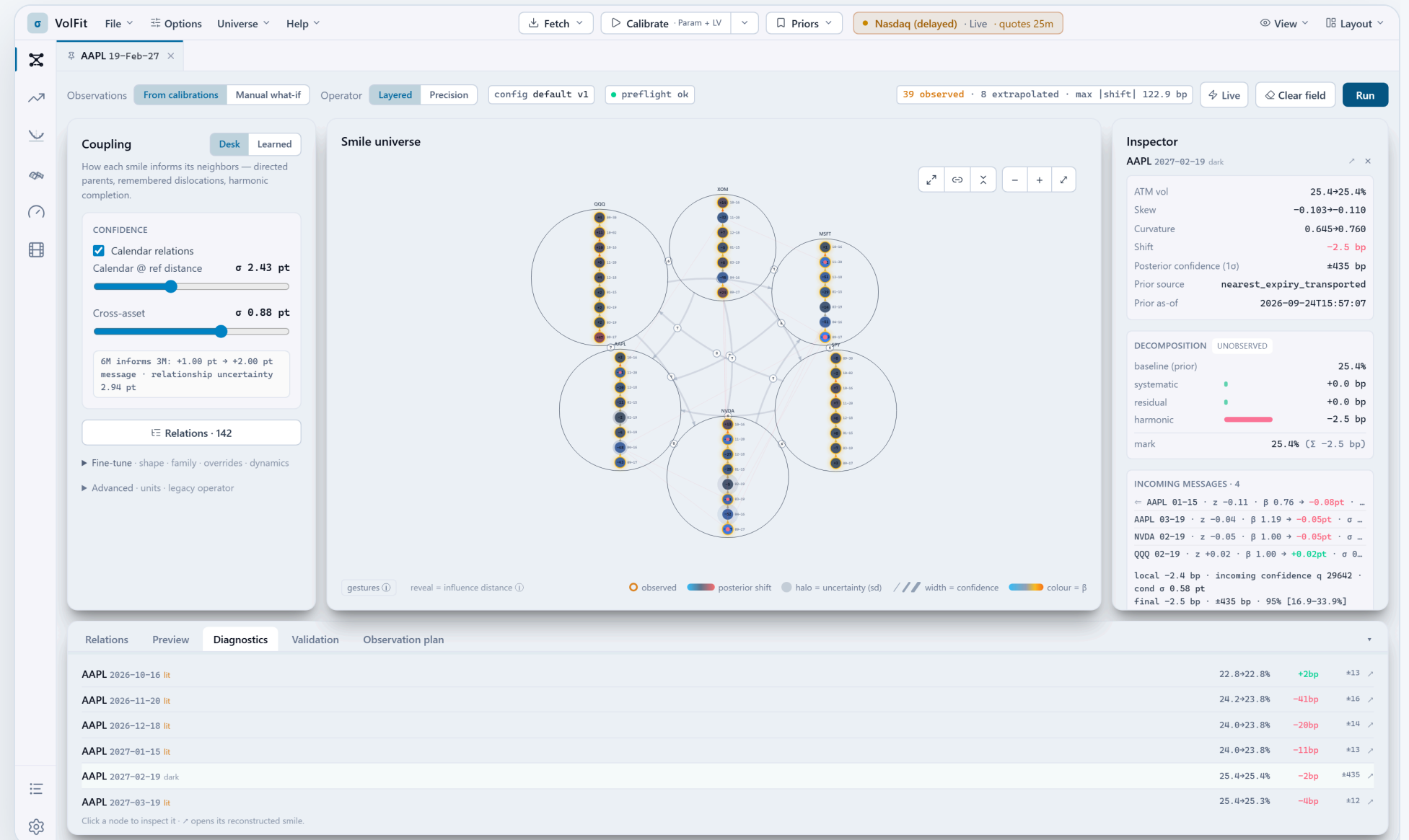
The graph estimates handle innovations across underlyings and maturities.

$$d_i = h_i^{\text{fit}} - h_i^0, \quad h_i^{\text{inferred}} = h_i^0 + \widehat{z}_i$$

- **Baseline**  $h^0$ : A saved prior resolved and transported to the current frame, with its source recorded.
- **Observed innovation**  $d$ : The change in the lit node's fitted ATM level, skew and curvature relative to that baseline.
- **Unknown innovation**  $z$ : The corresponding change at a dark node, inferred from configured relations and available observations.

## IN THE APP

Graph lens (Alt+1): Run, then click a dark node — the Inspector's Prior source, Transport, Innovation and Posterior confidence ( $1\sigma$ ).



**24 September capture:** 39 observed nodes and eight extrapolated nodes. Calendar and cross-asset relations link the selected universe.

# Graph operators and their assumptions

The operator determines how relations, observations and retained state contribute to an inferred move.

$$\hat{z}_F = -Q_{FF}^{-1}Q_{FS}d_S$$

- **Harmonic example:** The equation fixes observed nodes S and completes free nodes F using reciprocal relations, with no extra unary terms.
- **Additional information:** Directed predictions and baseline screens add target-side information. Source uncertainty still contributes to the output covariance.

| OPERATOR           | MATHEMATICAL CONSTRUCTION                                      | BEHAVIOUR                                                     |
|--------------------|----------------------------------------------------------------|---------------------------------------------------------------|
| Smooth field       | A quadratic smoothness prior plus baseline anchoring           | Nearby innovations are regularised together                   |
| Precision messages | Pairwise Gaussian relation factors                             | Jointly condition all connected innovations                   |
| Layered            | Directed predictions, residual state and reciprocal completion | One-way influence where specified; fresh certified boundaries |

# A precision-message relation

An edge specifies an expected response and the variance of departures from that response.

$$z_i = \beta_{ij} z_j + \epsilon_{ij}, \quad \epsilon_{ij} \sim \mathcal{N}(0, p_{ij}^{-1})$$

$$Q_{\text{msg}} = \sum_{j \rightarrow i} p_{ij} (e_i - \beta_{ij} e_j) (e_i - \beta_{ij} e_j)^\top$$

- $\beta$ : Maps an informer move into the receiver's units.
- $p = 1/\sigma^2_{\text{edge}}$ : Sets the relation precision. With a known source and no competing anchors, it changes uncertainty, not  $\beta$ .
- **Arrow semantics**: In the joint precision system, the factor couples both variables. Strictly one-way influence belongs to the layered operator.

| ILLUSTRATIVE SOURCE: 6M +1 VOL POINT | $\beta = T_{\text{source}}/T_{\text{receiver}}$ | PREDICTED MOVE |
|--------------------------------------|-------------------------------------------------|----------------|
| 3M receiver                          | 2                                               | +2 points      |
| 1Y receiver                          | 0.5                                             | +0.5 point     |

# Calendar and cross-asset relation controls

The relation editor exposes the response coefficient and uncertainty in the receiver's units.

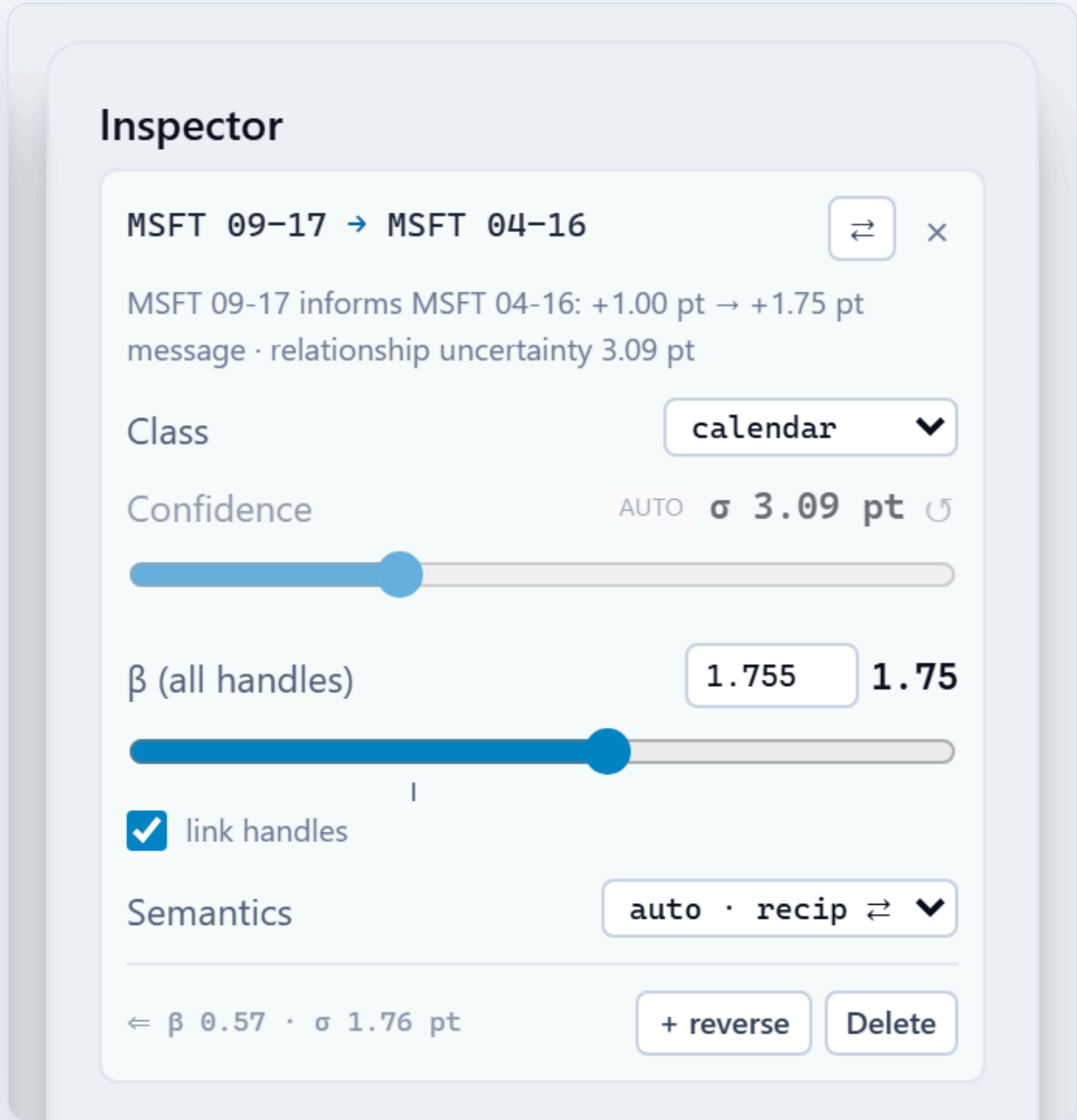
- **Calendar relation:** The automatic maturity rule supplies a response  $\beta$ ; the reference-distance control sets relation uncertainty.
- **Cross-asset relation:** An index, sector or selected peer can inform another name. Direction belongs to the selected operator's semantics.
- **Edit and inspect:** Change  $\beta$  or  $\sigma_{\text{edge}}$ , inspect the resulting vote, then compare the node's central move and marginal band.

### ILLUSTRATIVE CALCULATION

A +1-point informer innovation gives a +1.755-point relation prediction. Reading the same factor backwards divides both the response and the noise scale by 1.755.

#### IN THE APP

Click an arrow on the canvas: the relation card — Confidence  $\sigma$ ,  $\beta$  with link handles, Semantics; every edit stages a draft (Apply / Discard), Live previews the re-solve.



24 September MSFT relation:  $\beta = 1.755$  and  $\sigma_{\text{edge}} = 3.09$  vol points. The reverse reading is  $\beta \approx 0.57$  and  $\sigma \approx 1.76$  points.

# The joint posterior and shared information

Solve all coupled innovations together so that repeated routes from one source remain correlated.

$$Q^+ = Q_{\text{msg}} + D_{\kappa} + H^{\top} R_d H$$

---

$$Q^+ \hat{z} = H^{\top} R_d d, \quad \Sigma^+ = (Q^+)^{-1}$$

- **Observation precision**  $R_d$ : The innovation is a difference between a fitted state and a baseline; both contribute uncertainty.
- **Baseline precision**  $D_{\kappa}$ : An explicit zero-innovation anchor, when enabled.
- **Reported confidence**: Use the marginal variance  $\Sigma^+_{ii}$ . Incoming precision and the diagonal of  $Q^+$  are different quantities.

| ILLUSTRATIVE TRIANGLE: ONE NOISY SOURCE, TWO ROUTES | TARGET VARIANCE |
|-----------------------------------------------------|-----------------|
| Joint Gaussian calculation, all precisions $p$      | $5/(3p)$        |
| Incorrectly treating routes as independent          | $6/(5p)$        |

# Layered propagation and residual memory

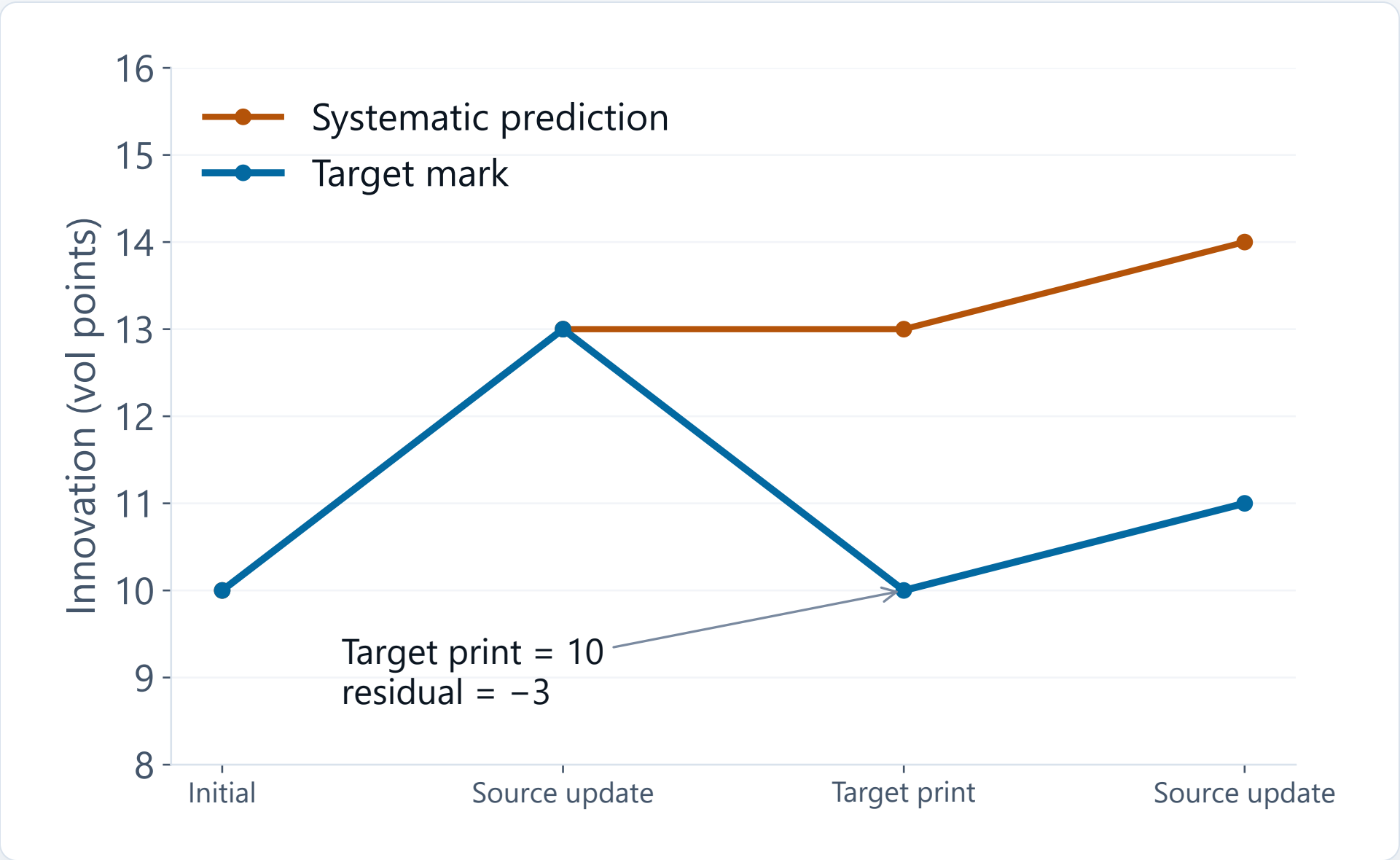
Directed predictions and reciprocal calendar completion serve different roles.

$$m_i^D = \sum_j \frac{p_{ij}}{q_i} \beta_{ij} m_j + m_{u,i}$$

---

$$u_{i,t+\Delta} = 2^{-\Delta/H_i} u_{i,t} + \omega_{i,t}$$

- **Directed layer:** A source move updates its receivers. A receiver's own print updates its residual, without revising the source.
- **Harmonic layer:** Reciprocal relations complete missing nodes around fresh certified boundaries and directed predictions.
- **Memory:** The residual mean decays with a chosen half-life; process noise widens its uncertainty between observations.



Illustrative  $\beta = 1$ , no residual decay: a +13-point systematic prediction meets a +10-point target print, storing  $u = -3$ . The next +14-point prediction gives +11.

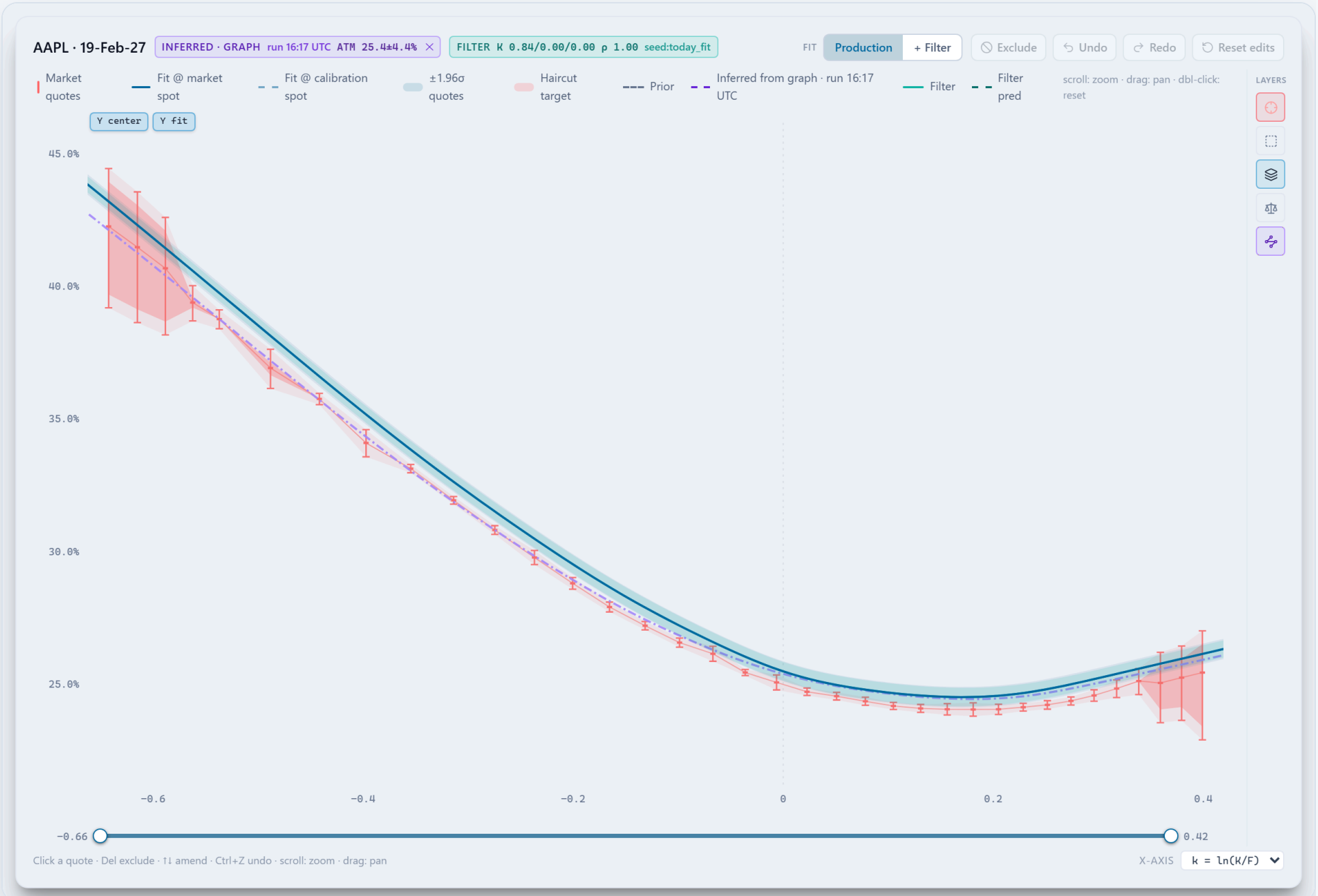
# Reconstructing and displaying an inferred smile

The graph supplies target handles; the baseline and model supply the remaining shape.

- **Reconstruction:** Map the inferred level, skew and curvature into the selected model around the transported baseline.
- **Inspection:** Compare target and achieved handles, the resulting smile and its uncertainty band.
- **Provenance:** Keep the inferred result distinct from observed calibration records. A graph-derived local-volatility surface is a subsequent projection.

### INTERPRETATION OF THIS CAPTURE

The reported ATM uncertainty is about 4.4 vol points, while the recorded graph shift is only a few vol bp. The band and central movement describe very different scales.



24 September AAPL February 2027 capture: violet is the inferred curve; the solid earlier fit is marked stale. Current quote bands are visible for inspection.

# Validation against a transported baseline

A held-out observation measures incremental information beyond the prior and spot-transport rule.

$$\zeta_i = \frac{h_i^{\text{held out}} - \hat{h}_i}{\sqrt{V_i^{\text{pred}} + V_i^{\text{held out}}}}$$

- **Leave one out:** Remove a node's current observation, infer it from the remaining information, then score its withheld calibration.
- **Baseline comparison:** Use the same prior, dynamics, as-of time and held-out set for the transported-prior result.
- **Uncertainty assessment:** Examine bias, standardized residual spread and interval coverage as well as mean error.

IN THE APP

Graph drawer ▶ Validation ▶ Compare operators (LOO) — start it while talking; RMSE · ζ mean · ζ std · cov 80 · cov 95 per operator.

RelationsPreviewDiagnosticsValidationObservation plan

🔬 Compare operators (LOO)

📄 offline benchmark artifact

|           | Transported prior | Smooth field | Messages |
|-----------|-------------------|--------------|----------|
| RMSE (bp) | 38.7              | 36.7         | 38.5     |
| ζ mean    | —                 | −0.02        | 0.06     |
| ζ std     | —                 | 0.11         | 0.72     |
| cov 80    | —                 | 100%         | 90%      |
| cov 95    | —                 | 100%         | 95%      |
| n scored  | 39                | 39           | 39       |

Same day, same universe, same knobs — only the operator differs. The transported-prior column has no calibrated uncertainty (ζ/cov n/a). The multi-regime verdict lives in the offline artifact.

24 September snapshot check, 39 held-out nodes: ATM RMSE 38.7 bp for the transported prior, 36.7 for smooth field, 38.5 for messages.

# Choosing the next observation

A candidate quote can be ranked by the reduction it would make in posterior uncertainty.

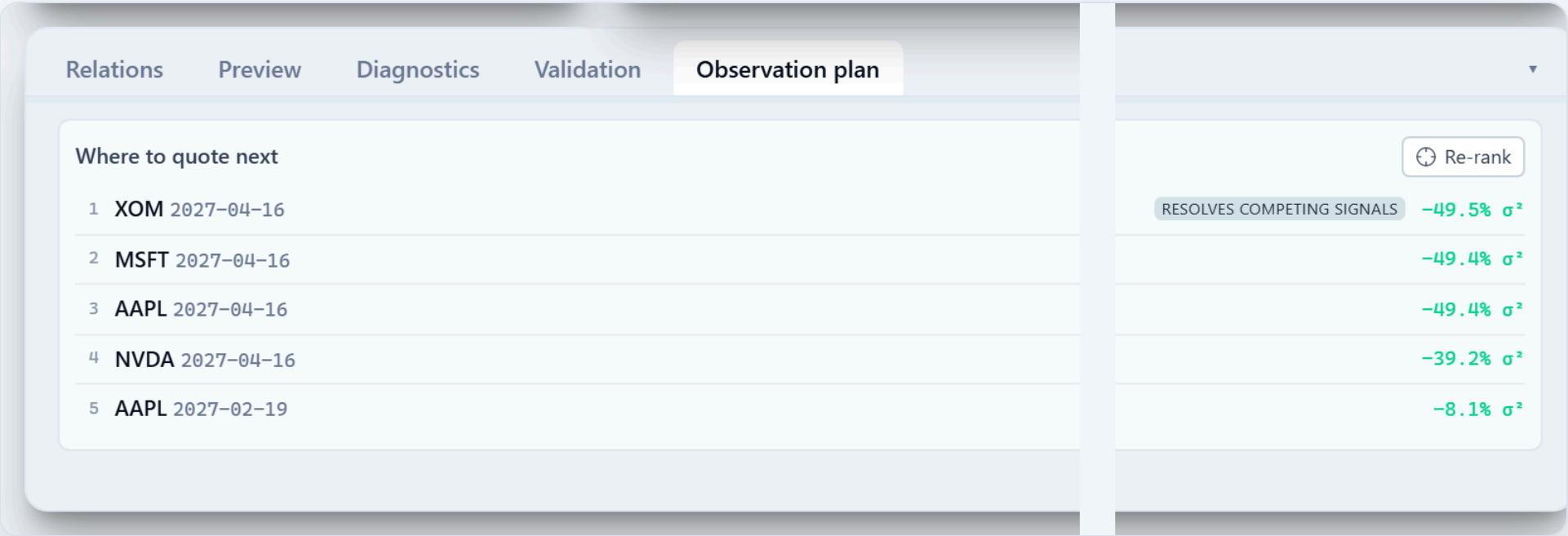
$$\Sigma_{\text{new}} = \Sigma - \frac{\Sigma a a^\top \Sigma}{a^\top \Sigma a + r}$$

- **Candidate measurement:** The row  $a$  selects a handle or linear combination;  $r$  is the expected observation variance.
- **Reduction:** The rank-one update shows which other nodes would also become better determined.
- **Ranking:** Compare the selected variance-reduction objective across candidate dark nodes. The result depends on graph and noise assumptions.

| RANK | DARK NODE          | VARIANCE REDUCTION                             |
|------|--------------------|------------------------------------------------|
| 1    | XOM April 2027     | −49.5% $\sigma^2$ · resolves competing signals |
| 2    | MSFT April 2027    | −49.4% $\sigma^2$                              |
| 3    | AAPL April 2027    | −49.4% $\sigma^2$                              |
| 4    | NVDA April 2027    | −39.2% $\sigma^2$                              |
| 5    | AAPL February 2027 | −8.1% $\sigma^2$                               |

IN THE APP

Graph drawer ▶ Observation plan: Where to quote next, the ranked dark nodes; Re-rank after changing a relation.



24 September capture: XOM April 2027 was ranked first under that run's covariance and candidate-observation assumptions.

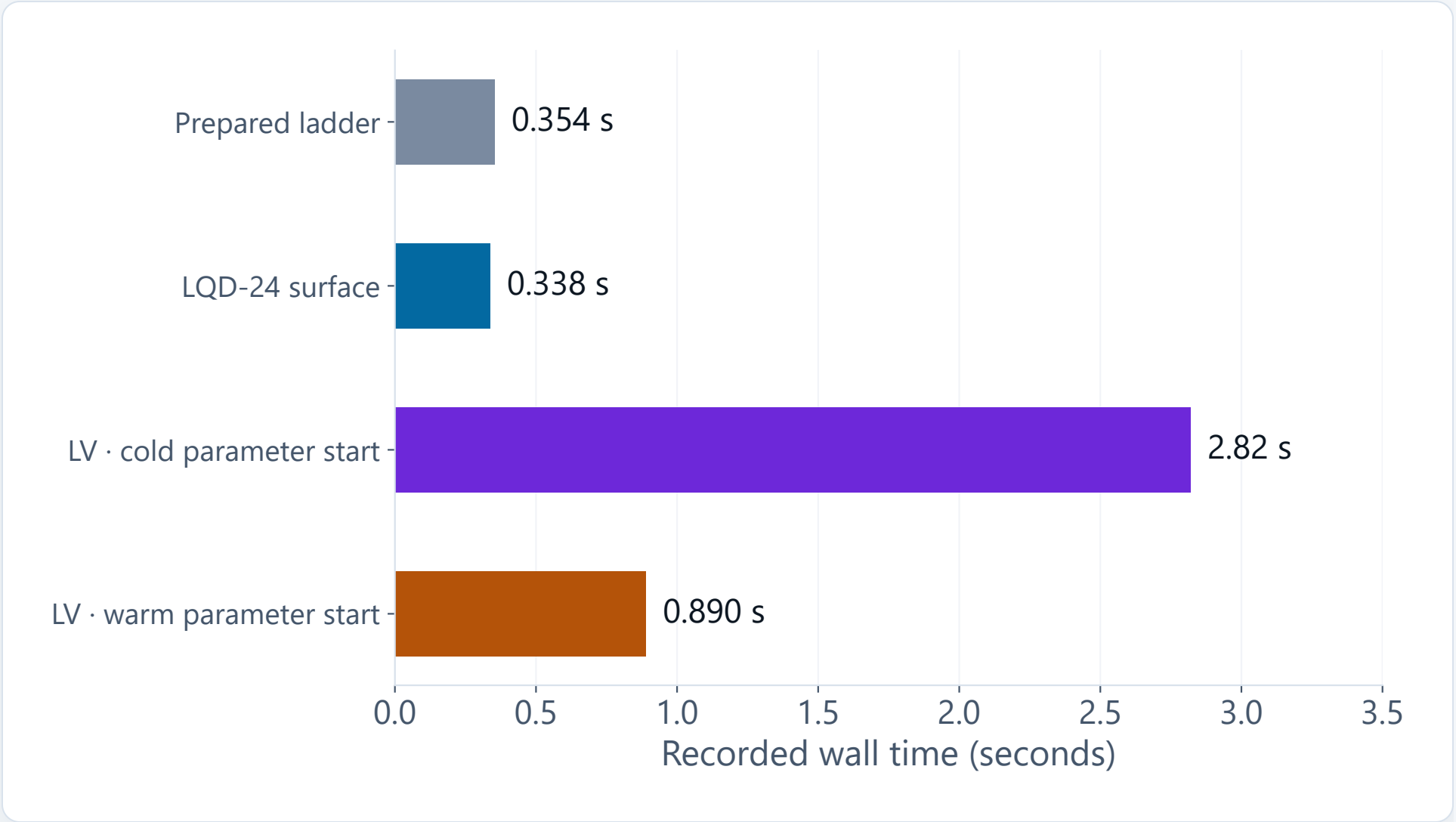
# Where calibration time is spent

Separate data retrieval, quote preparation, parametric fitting and the local-volatility inverse problem.

- **Measured stages:** Prepared ladder: 0.354 s. LQD-24 surface: 0.338 s. LV: 2.82 s from a cold start, 0.89 s from a warm parameter start.
- **Different reuse:** A warm parameter seed, a compiled-kernel cache and cached prepared quotes save different work.
- **Parallelism:** The calibration pool distributes ticker groups. It does not divide one local-volatility surface across workers.

RECORDED SCOPE

Mid target on one captured SPY ladder. The whole Calibrate job was 3.34 s in its own serial run. Separately timed stage medians need not sum to that job measurement.



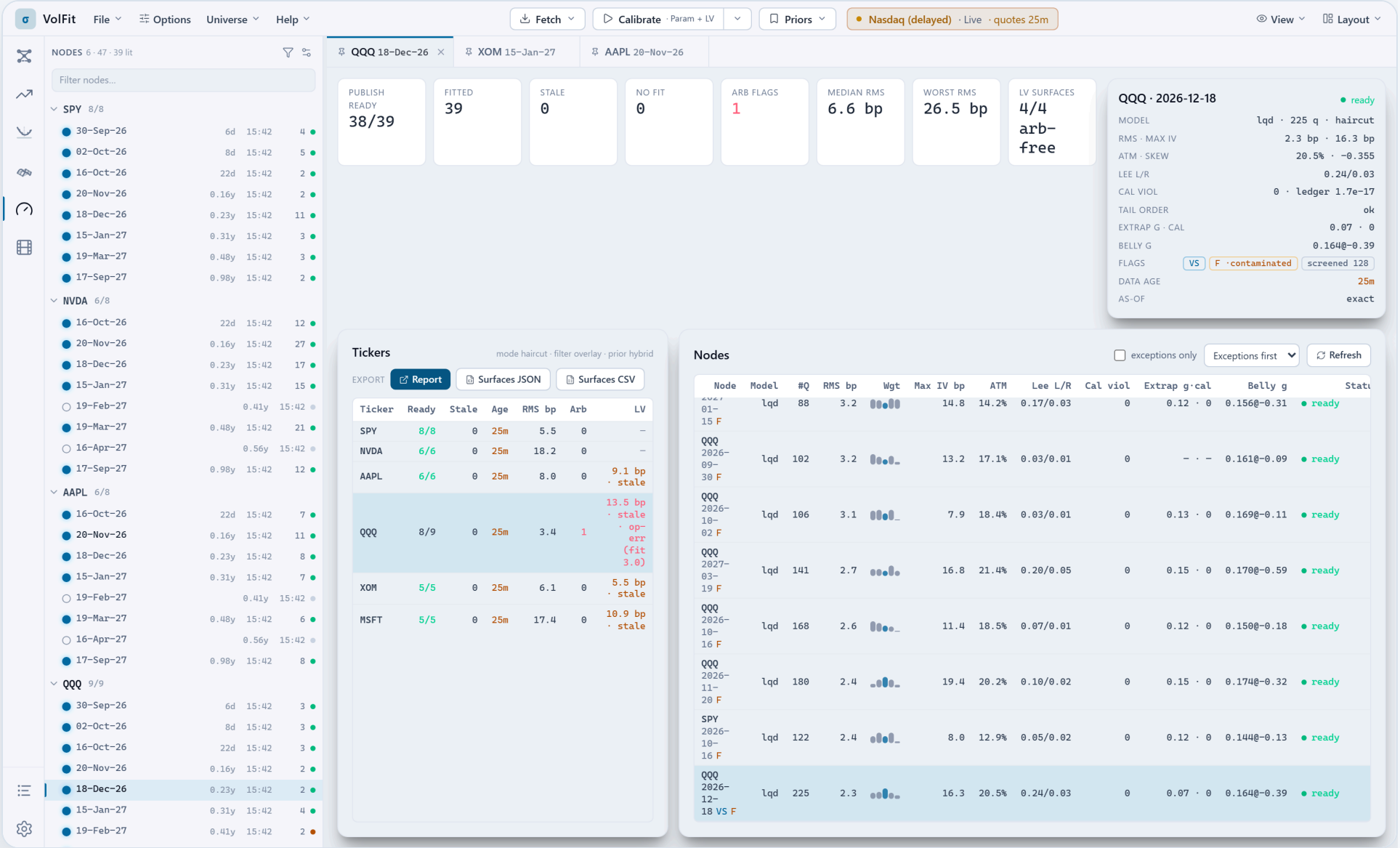
Recorded 23 September SPY experiment: nine expiries, 875 fit quotes, Intel i7-12700H. Local volatility used 253 vertices and a 368 × 116 pricing lattice.

# Reading fit quality and publication state

Fit error, data freshness, arbitrage diagnostics and inference uncertainty describe different properties.

- **Fit error:** Read the selected target, weights, quote count and maximum error alongside RMS.
- **Consistency:** Inspect density, adjacent-expiry order, tail policy and the refined local-volatility reprice where relevant.
- **State:** Check data age, fit timestamp, source, stale status and whether the displayed result is fitted, transported or inferred.

**IN THE APP** Quality lens (Alt+5): the tiles, the per-ticker rollup and the node table sorted Exceptions first; a node card's certificates and data age.



24 September capture: the dashboard aggregates cached fit records. Its displayed readiness is a result of the configured checks at that time.

# A complete worked sequence

Follow one liquid node and one withheld expiry through the calculation.

| ACTION                         | INSPECT                                            | CALCULATION BEING ILLUSTRATED           |
|--------------------------------|----------------------------------------------------|-----------------------------------------|
| Select the node and source     | Timestamp, exercise style, forward and dividends   | Definition of the market input          |
| Fit one smile                  | Quote bands, error distribution, density and tails | Model and objective                     |
| Inspect the expiry ladder      | Total variance, event clock and calendar gap       | Maturity consistency                    |
| Move the spot                  | Stored fit and transported curve                   | Chosen spot–vol response                |
| Compare anchoring modes        | Active prior rows, filter gain and innovation      | Use of earlier observations             |
| Withhold a current observation | Baseline, incoming relations and inferred curve    | Graph inference and reconstruction      |
| Replay and score               | Same frames, independent lanes, held-out errors    | Sensitivity to the selected assumptions |

C L O S E

# One calculation, read in dependency order

Observed quotes, fitted quantities and inferred quantities, with the assumptions that separate them stated where they apply.

- **Observed:** Quotes brought to one forward, one exercise convention and one time stamp before any model sees them.
- **Fitted:** A smile or a local-variance surface, read together with its objective, its weights, its tail policy and its maturity coupling.
- **Inferred:** Transported, persisted, filtered or graph-inferred marks, each with its baseline, its provenance and its uncertainty.
- **Dated:** Timings and scores belong to the recorded experiment that produced them; the captures are the 24 September session.